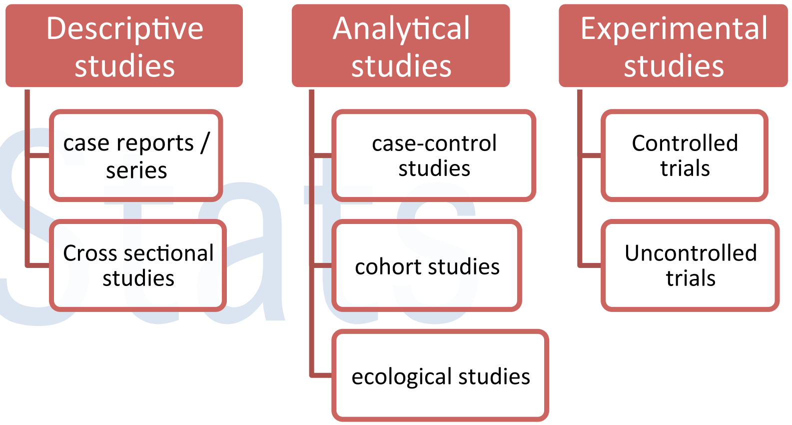
Study designs

The design of any research study can be described on the basis of two aspects related to the recording of exposure and outcome:

1. **The identification (or grouping) of study subjects:** i.e. selection of subjects on the basis of differences in exposure (exposed vs. non-exposed) or differences in outcome (cases vs. controls)
2. **The time course of investigation** i.e. whether the study looks forward (prospective) from the exposure to the outcome, or backwards (retrospective) from the outcome to the exposure, or static (cross-sectional) where the exposure and the outcome are measured together.



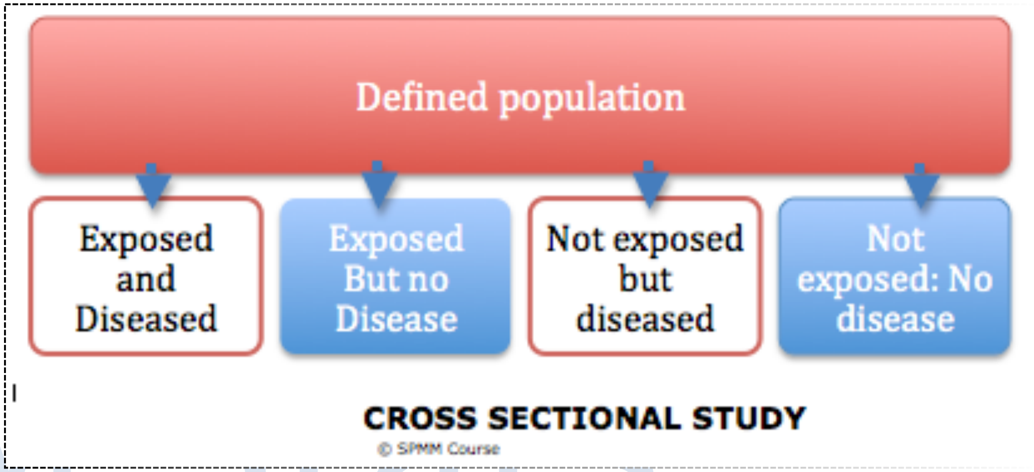
In **prospective research,** subjects exposed to a risk factor or intervention are assembled and then the researcher waits for the outcome to take place. In **retrospective research**, a group of individuals has been assembled because they have already experienced the exposure in question.

<u>**Descriptive studies**</u> aim to describe the characteristics of a group of subjects. There is no testing of a causal hypothesis or any comparisons with other groups. Descriptive studies describe the characteristics present in members of the study group and discuss the distribution of these characteristics.

The simplest forms of descriptive studies are **case reports** and **case series**.
A special type of descriptive study is a cross-sectional study. **Cross-sectional studies** can *suggest* the presence of putative relationships among variables. They are also called **prevalence studies** and are used to describe various attributes of the subjects in a study group but *at a single point in time (a snapshot).* **Surveys** are cross-sectional studies. **Census** is a special cross-sectional study where instead of a subset, the sample includes the whole of the population. In **prospective longitudinal studies,** observations are

repeated in the same population over a prolonged time. This is helpful to study the natural course of illness, study risk factors and find out incidence rates. But longitudinal studies are costly and time consuming.

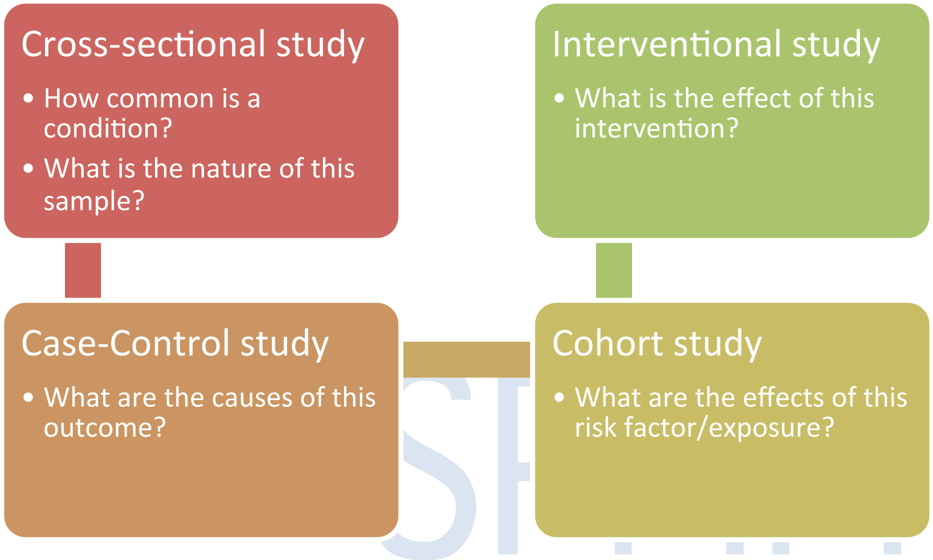
Analytical studies involve a comparative analysis of two groups to measure the association between variables of interest (e.g. a disease and a risk factor / an exposure and an outcome). The common types are case-control studies and cohort studies.
In a **case-control study,** the researcher assembles a group of subjects, some of whom have already experienced an outcome ("cases") and others who have not ("controls"). Then, the researcher estimates the proportion in each group who have experienced the exposure of interest. The hypothesis of a putative causal relationship is supported if the proportion of exposed subjects is larger among the cases than in the "controls".



A **cohort study** is an observational study of a group of people with specified characteristics or exposure who are followed over a period of time to detect events/outcome.

A special type of analytical study is **ecological study**. Here the data are gathered to describe what happens in a group rather than the individual. Ecologic studies consider populations rather than individuals as the unit of measurement – thus there is an assumption that all members of a given population ar equally exposed and carry the same risk of a disease, disregarding within-group differences. For example, in countries with low religious attachment rates, rates of suicide are high. Concluding that low religious attachment is related to high suicide may not be valid, and it tells nothing about the individuals who commit suicide; those individuals who commit suicide may or may not be religious. Drawing subject-level conclusions based on group-level observations in an ecological study is termed as *ecological fallacy*.

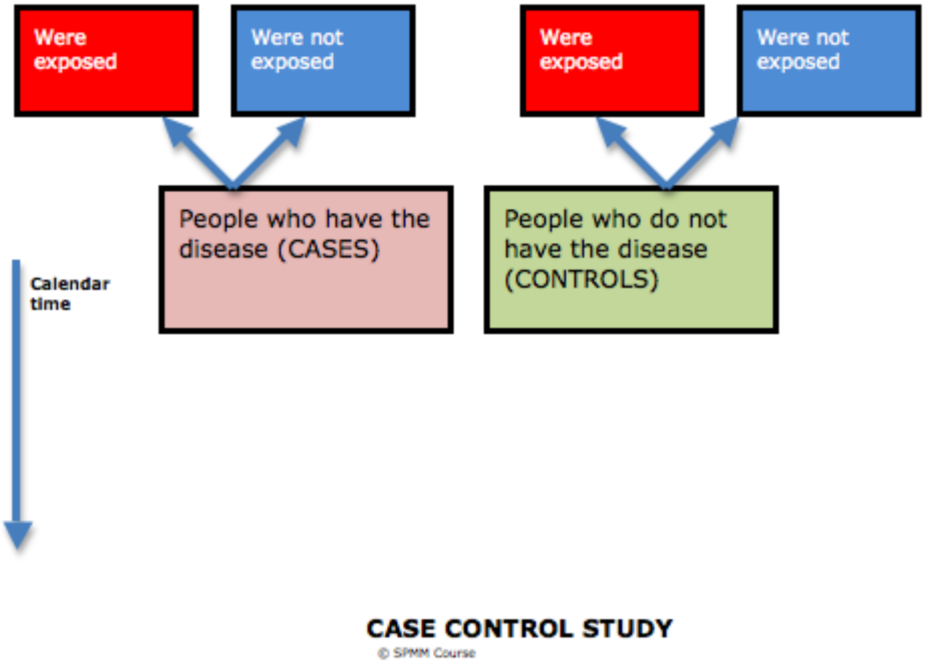
<u>Experimental studies</u> are generally prospective cohort studies where the exposure is experimentally assigned to the groups in a variable fashion to observe the effect. Unlike the cohort studies, here deliberate manipulation of exposure (intervention) takes place. **Clinical trials** are experimental studies. They are also called **interventional studies**.



Clinical enquiry	Best study design to answer the query
Treatment effectiveness	Pragmatic RCT (or systematic reviews)
Treatment efficacy	Experimental RCT (or systematic reviews)
Causation (aetiology)	Cohort or case-control
Prognosis	Cohort
Diagnostic assessment (evaluating a new tool)	Cross-sectional comparison to gold standard
Health economics	Cost-effectiveness study
'meaning' or health 'experience.'	Qualitative studies

Case control design

- This is a retrospective design; the research study starts from defined groups of cases and controls – i.e. separating diseased and non-diseased subjects. Then retrospective data on exposure to putative risk factors is collected.
- Both exposure and disease have occurred before the onset of the study.
- It is often the first mode of studying a suspected association of causality.
- It is mostly useful in diseases where exposure is common, but the disease occurs only in a small proportion of those exposed.



- Advantages:
  - Easy to carry out
  - Less time-consuming
  - Less expensive
  - Suitable for investigating rare diseases
  - Subjects are not exposed to any new risks
  - Several different etiological factors for single disease can be studied
  - No attrition problems
- Disadvantages:
  - Highly prone to selection and recall bias
  - Control group selection may be difficult
  - Incidence cannot be measured - so odds ratio only – not relative risk – can be measured
  - Cannot prove causality
  - Temporality is difficult to determine retrospectively.
- Defining a case is crucial – strict diagnostic

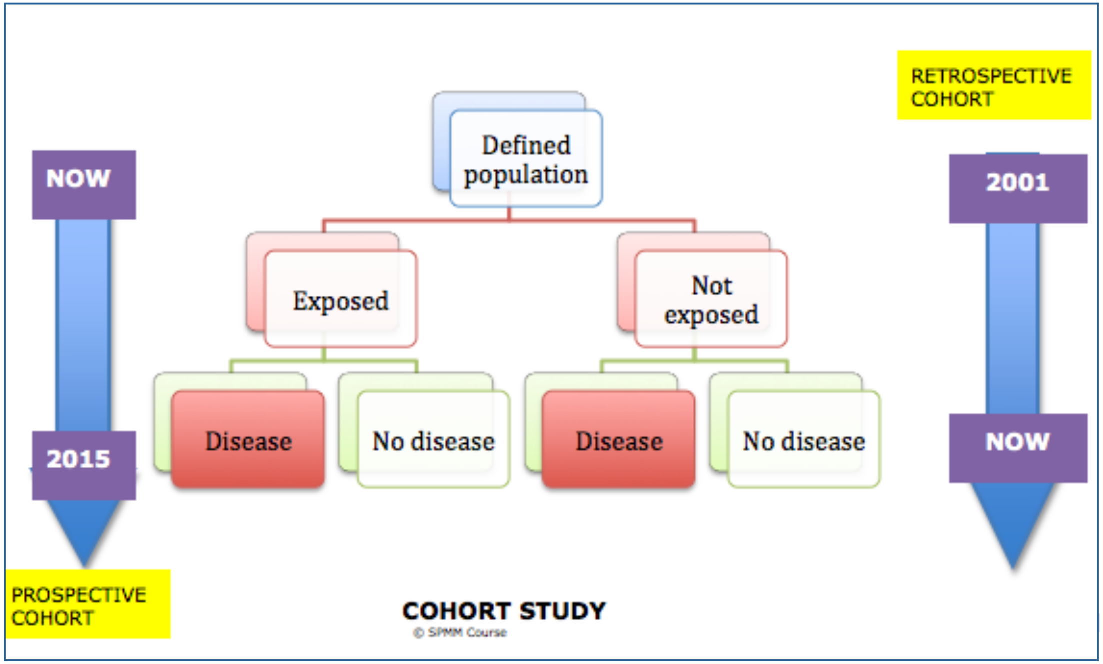
criteria and eligibility criteria must be followed.

- Cases can be obtained from a hospital or general population according to the requirement. Controls by definition must be identical to cases except for the presence or absence of disease and **recruited independently of exposure status**.
- Controls can be hospital patients who have a different health problem to the one studied, relatives of patients studied, neighbourhood controls or general population controls.
- Up to a ratio of 4:1, when the number of controls is increased, the power of a study increases. But not after exceeding this ratio.
- To ensure comparability the cases and controls are often **matched** – i.e. certain selected variables such as age, sex, etc. are matched when controls and patients are recruited. **Matching** can be done in groupwise or pairwise fashion, the latter being more time consuming.
- Overzealous matching may not only make recruitment of controls difficult, but can also reduce the risk difference between the two groups. Overzealous matching may inadvertently result in matching for putative causal factors as well.
- To estimate the risk of exposure to case-control studies a cross-product ratio called **odds ratio** is used. This is because relative risk cannot be calculated as incidence rates are not available (no longitudinal observation to see new cases developing as in cohort studies – see below).
- Note that if a disease is very rare (prevalence < 5% arbitrarily), then odds ratio approximates the relative risk.
- OR can have values only between 0 and infinity (no negative values). Hence to calculate confidence intervals one needs to use log transformations.

Cohort study:

- A cohort is a group of persons sharing a common aspect e.g. **birth cohort** – all those who were born on September 13, 1981.
- Similarly, **exposure cohort** refers to all those who were exposed to a risk factor.
- **Inception cohort** refers to a group of patients who are assembled at a single (or narrow) point of time based on a common factor.
- In a cohort study, exposure cohorts are followed up in parallel with a non-exposed group, to detect the development of a disease.
- Here, exposure has occurred, but the disease is yet to occur when the study starts.
- It is especially useful when exposure is rare, but the likelihood of disease is notably high after the exposure.

❖ Cohort studies are only possible if an easily obtainable, cooperative and stable cohort that can be followed up as needed is available. The non-exposed control cohort must be comparable to the study cohort in all aspects except the exposure; they must not have the disease at the time of inception into the study. The control cohort may be **internal control** or **external control**. Internal controls are subgroups in exposure cohort according to dose or degree of exposure e.g. ½ pack a day smokers vs. 1 pack a day smokers. Though generally almost all cohort studies are prospective it is possible to do a **retrospective cohort** by identifying a cohort using records and then tracing them forwards to see if they have developed a disease of interest. E.g. consider military recruits in the year 1940 who got physically disfigured (exposure), trace them till year 1998 and observe health records for development of PTSD, comparing them to recruits in the same year who did not suffer a disfigurement. (see the table below for differences between retrospective, prospective cohort and case-control). Combining retrospective and prospective method is also possible, the former used to identify a cohort, which is then prospectively followed up.



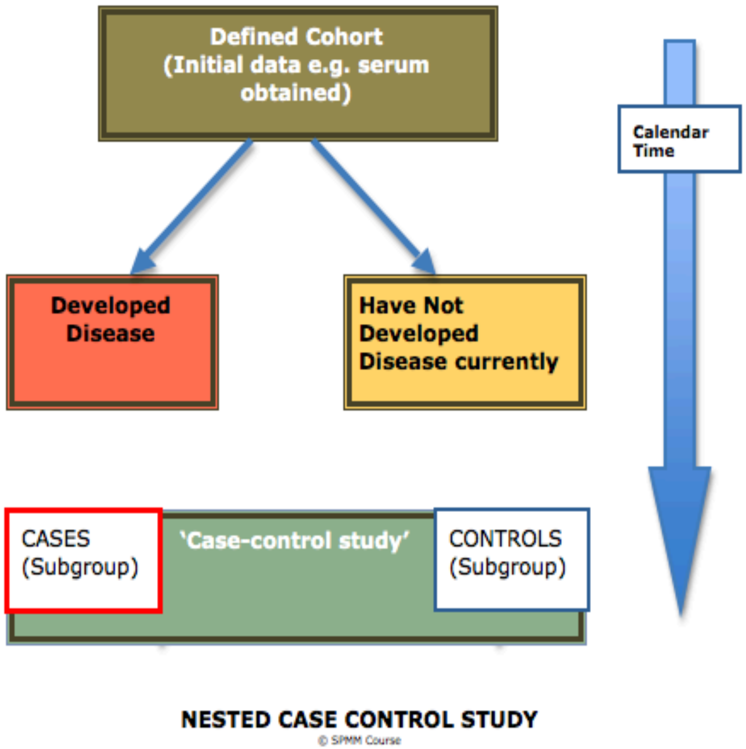
❖ In cohort studies, the **relative risk** can be calculated as the ratio of the disease (outcome) in the exposed to the disease in non-exposed. Odds Ratio can also be calculated from cohort studies.

Study design	Subjects identified according to	Determination time
Cross-sectional	Neither exposure nor outcome	Current estimate of exposure and outcome
Case-control	Disease (outcome) status	Past estimate of exposure; current estimate of outcome
Prospective cohort	Exposure status	Current estimate of exposure; future estimate of outcome
Retrospective cohort	Exposure status	Past estimate of exposure; current or past estimate of outcome

COHORT STUDY: Disadvantages	Advantages
1. Difficult to carry out as more time/resource consuming 2. Not suitable for investigating rare diseases- thousands of exposed persons must be observed in this case. 3. Only one etiological factor can be studied at a time. **4.** Attrition / drop-out is a major issue	1. Less prone to selection and recall bias 2. Incidence can be measured - so relative risk can be measured 3. Causal association can be strongly supported compared to case-control design 4. Temporality is established easily 5. Multiple effects of a single exposure could be observed. 6. Dose-response relationships could be calculated **7.** The natural course of exposure to disease pathway can be studied in addition.

From Silman & Macfarlane, Epidemiology: A practical guide. 2nd ed. Cambridge University Press.

Nested case-control study: A study in which both the cases and controls are drawn from within a cohort, rather than including the entire cohort population in the study. Such studies can be conducted at a fraction of the cost and yet produce the same findings with nearly the same level of precision. Case-control studies can also be nested within other study designs such as cross-sectional studies though this is less common.



Interventional Studies

These are either RCTs (randomized controlled trials) or uncontrolled trials. The major features of RCTs differentiating it from other types of studies are

1. **Experimental intervention** – one group remains control (or standard) and the other(s) receives some intervention
2. **Randomization** occurs – i.e. after a subject enters the study he/she is randomly allocated to either control arm or intervention arm (or other arms) through a process of randomization.
   a. This may be **simple randomization** e.g. randomizing each entrant separately using a computer generated list; or
   b. **Block randomization** – where randomization occurs in a set of specified numbers, say blocks of 6 etc., to ensure equal number distribution between two arms. Larger the block size, lesser the ability to predict allocation.
   c. In **stratified randomization** baseline characteristics which may have an implication on final outcome are stratified so that equal distribution across groups is enforced. The gain in precision through stratification is minimal for large studies. Stratification does not eliminate the need for adjustment for differences in the baseline composition of the study groups if found.
   d. **Cluster randomization** – where unit of randomization and analysis is various centers or catchment-zones and not individual patients – so after a cluster is allocated to an arm, all individuals in the cluster receive same intervention (or non-intervention). (See ***minimisation*** below). Within a cluster, patients are more likely to respond in a similar manner, and thus can no longer be assumed to act as independent units. This lack of independence in turn leads to a loss of statistical power in comparison with a 'patient' randomized trial. As a result, in cluster RCTs, *effective sample size* *(ESS)* is used in place of actual sample size when undertaking power calculations. To calculate ESS, statisticians use a measure called *intracluster correlation*, which represents the degree to which the various individuals in a cluster resemble each other in the outcome measure.
   e. **Minimisation:** this is useful in two situations.
      i. You want to do a stratified randomisation as you are worried about unequal distribution some very important covariates. But your trial is too small with too few participants for stratified randomisation.
      ii. You are conducting a cluster randomisation and units of randomisation being clusters you end up with small numbers. You want to minimise gross differences in confounder (covariant) distribution.

Characters of sound randomisation

- Reproducible order of assignment (not possible for coin-flips though this method adequately generates a random sequence)
- Documented methods for assignments
- Assignments remain concealed (see below)
- Future assignments not predictable from past assignments
- Ability to detect departures from established procedures

In the above situations, we can use minimisation to achieve a balance between treatment groups. In minimisation schemes, the next allocation depends on characteristics of those already allocated. Allocation of each participant aims to ensure a balance of prognostic factors between groups. The disadvantage is that this method is inferior to proper randomisation as it allocation is somewhat exposed and 'controlled' manually; in fact only the first allocation is truly random. Using computerized programs can reduce bias while minimizing.

f. **Quasi-randomisation** refers to 'randomizing' using even/odd numbers of the date of birth, day of the week patient was seen, etc. These are not reproducible methods, and the sequences cannot ensure equal distribution of variables. These must be avoided.

g. Uses of randomisation
   i. It permits use of probability theory in making inferences
   ii. Eliminates effects of bias
   iii. Facilitates blinding
   iv. Distributes baseline characters in an unpredictable fashion

3. Randomisation improves **generalizability,** but it does not ensure the results will always be precise –**precision** depends very much on the sample size. Having a large sample size does not mean randomization is unnecessary.

Special types of RCTs:

- **Crossover trial** – this refers to an interchange of study and control groups after a washout period so that all subjects in a study receive both placebo and treatment but only in a different order. This is an economic design as number of required subjects is lesser. But this cannot be used for diseases that are cured by the use of medications. **Parallel RCT** is a term used to denote the common form of RCT, which is the opposite idea of crossover trials – i.e. both control and intervention happens in a parallel, not serial fashion as in crossover trials.

Factorial design		Naltrexone	
		YES	NO
Cognitive coping skills therapy (CCT)	YES	Both Naltrexone & CCT	CCT only (placebo)
	NO	Naltrexone only (supported)	Placebo only (supported)

- **N of 1 trial** – here a single subject is administered placebo and active intervention (or two different interventions) in tandem under double-blind controlled conditions. This is the best way of optimizing a treatment schedule for an individual patient – but results cannot be generalized to other patients.
- **Factorial trials -** Most RCTs evaluate a single therapeutic factor of interest, often a novel intervention that is compared with one or more alternatives, or a placebo. In a trial using a 2x2 factorial design, participants are allocated to one of the four possible

combinations (2 treatments, 2 levels). A 2x2x2 trial indicates 3 treatments at 2 levels. For example in a 2x2 factorial RCT participants would be allocated to drug A alone, drug B alone, both drug A &B, placebo. In this way, it is possible to test both the independent effect of each drug and the combined effect. Consider the design (shown in the table) from **Heinala P, et al. (2001) Targeted use of naltrexone without prior detoxification in the treatment of alcohol dependence: a factorial double-blind, placebo-controlled trial. J Clin Psychopharmacol *21*: 287–292.** For the same question if a multi-arm trial is conducted, the cost may be more but power reduced. Factorial trials are difficult to operationalise.

- **Patient preference trials:** When evaluating certain interventions some patients may have a strong preference for a particular therapy. They are then allocated the therapy they prefer. Only those with equivocal or no preference are then randomised. Note that the primary analysis should still be limited to randomised groups. But the researcher can obtain data from an observational cohort (preference group) in a parallel fashion. This maximises recruitment in a trial. One can also analyse the effect of motivation on intervention effectiveness. This makes the results more generalisable. But obtaining consent for such trials may be complex and recruiting those with *no preference* may be hard. (**BMJ 1998; 316: 360)**
- **Zelen's modified RCTs**: Generally informed consent is obtained before randomising a sample for RCT. But the problem with this standard procedure is that a substantial number of patients may withdraw if they suspect they are not getting the treatment they hope for. Their compliance may become poor if they have a hunch on what intervention they are getting. Zelen's modification suggests randomisation before obtaining consent. The consent is sought then from those randomised to experimental treatment. This approach is controversial with respect to ethics. MRC does not accept this design as ethically valid. **(BMJ 1998; 316: 606)**

- **Non-inferiority trials:** Most RCTs are conducted to demonstrate the superiority of one intervention over the other. These can be called as superiority trials. In some instances, especially if a well-established standard treatment is available, drug companies may want only to demonstrate that the new intervention is 'not inferior' to standard intervention. This type of RCT is called as an equivalence trial (same efficacy as the comparison) or non-inferiority (not significantly worse than the comparison by more than a certain 'allowed' margin) trial. Note that the null hypothesis for such trials must be 'there is a significant difference between two interventions' and not otherwise as seen in superiority trials. Also, a higher sample size is required for an equivalence or non-inferiority trial compared to a superiority trial at the same level of power and significance**.** *(In fact theoretically true equivalence can only be demonstrated if the sample size is infinity as the effect size, by definition, is zero in such studies!)* Also intention to treat analysis may blur actual differences between the groups and, in fact, may support the claim of equivalence. Hence, it is best to report both ITT and per-protocol analysis in these trials. **(BMJ 1996: 313; 36–39).**

Other types of interventional trials:

- **Uncontrolled trials** – here only one group is observed for effects of an intervention – there is no control group. The study inference is made using a historical control i.e.the status of the treated group before intervention
- **Before and after trials** – here, the outcome is assessed before and after some intervention in one group of subjects. The observed change may be in fact attributable to a factor other than the intervention itself, making this design inherently weak. There is no randomization in this design.
- **Multi Arm trial:** This is a simple extension of RCT where more than one study arm is included. It is easy to design and allows intervention-1 vs. intervention-2 comparison in addition to intervention-1/2 vs. placebo effect. But this requires a larger sample size and power may be reduced as a consequence.

Pragmatic RCTs:

- **Efficacy** refers to how well a drug can work under optimal circumstances.
- **Effectiveness** refers to how well it works under usual practice circumstances.
- **Efficiency** measures whether health care resources are being used to maximize value for money

The fundamental differences between efficacy trials and effectiveness trials relate to objectives and motivation for the trial. An efficacy trial is undertaken to meet regulatory approval. An effectiveness trial is designed to convince formularies and payers of the actual usefulness of the drug in current practice. Such effectiveness trials for which the hypothesis and study design are developed specifically to answer the questions faced by decision makers are also called **pragmatic or practical clinical trials (PCTs),** e.g., CATIE was a pragmatic RCT. One need not model all details of intervention (such as when to escalate the dose, etc.) beforehand in pragmatic trials – these are attended to as per the need of the patients in trials.

Unnatural circumstances in a RCT compared to regular clinical practice
Recruitment: from specialist centers
Patients with co-morbid medical or psychiatric disorders are excluded
Patients are carefully selected to generate homogeneous diagnostic groups
Patients are allocated the treatment at random (no negotiation during prescription)
Patients are given detailed information for informed consent
Placebo is used to compare active treatment
Patients are followed at frequent intervals
Patients get detailed checklists of side-effects
Assessment endpoint is typically 4–6 weeks
Often symptoms measures are the outcomes considered.
Patient and clinician are blind to treatment.

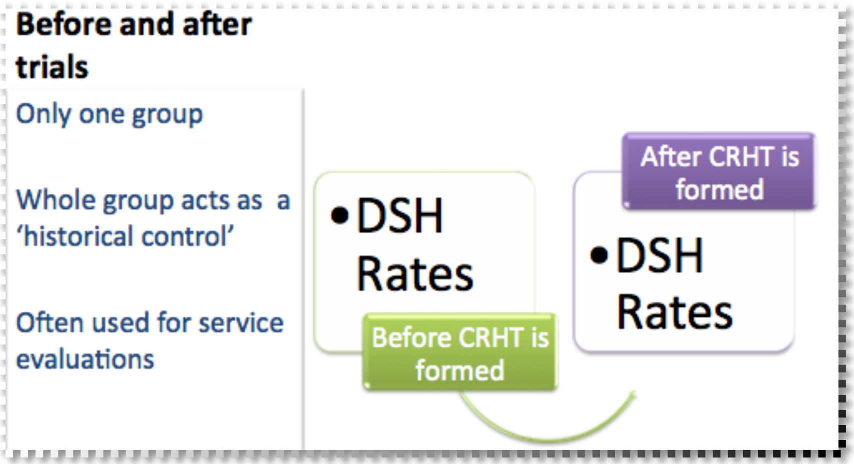
The characteristic features of pragmatic RCTs are that they

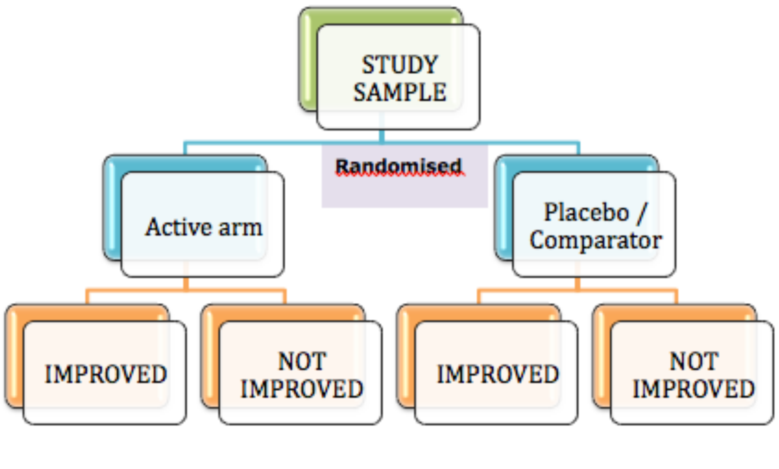
(1) Select clinically relevant alternative interventions to compare,
(2) Include a diverse population of study participants,
(3) Recruit participants from heterogeneous practice settings, and
(4) Collect data on a broad range of health outcomes.
(5) They do not tell how well a drug works (EFFICACY) – they inform how EFFECTIVE is the intervention in the real world.

Pragmatic RCTs
• Reflect the heterogeneity of patients in clinical practice
• Minimise exclusion criteria
• Focus on groups with a wide range of diagnoses
• Define patient groups by presentation rather than diagnosis
• May not employ placebos
• May not be blinded
• Must carefully conceal allocation during randomisation
• May evaluate real world, function based outcomes

Problems with pragmatic designs include -

1) Substantial drop-outs can occur
2) Blinding can fail
3) Less rigorous methodology
4) Higher rates of non-specific treatment effects.





PARALLEL RCT

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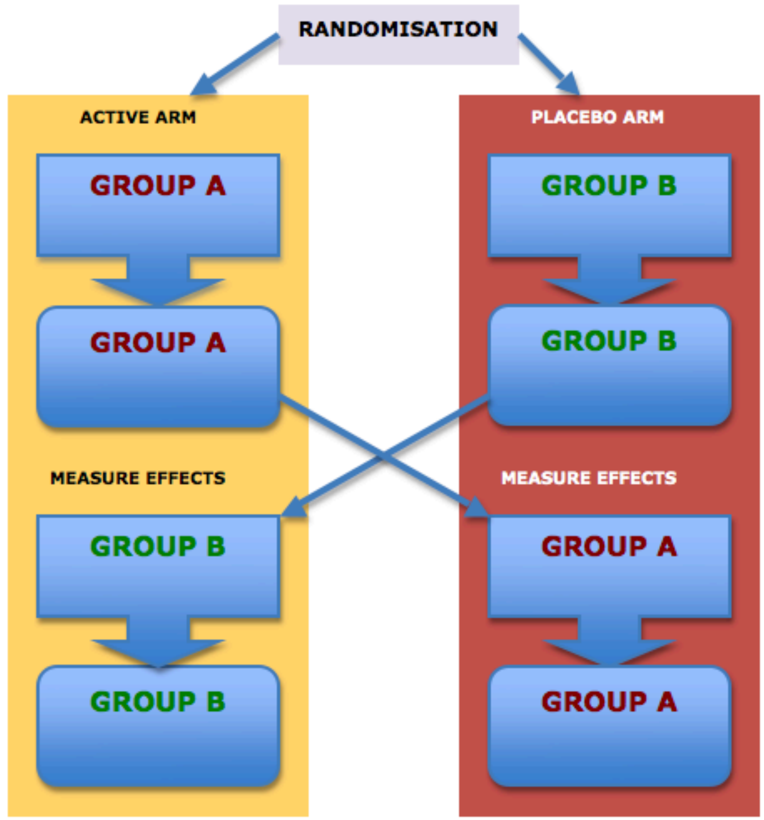

CROSSOVER RCT

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Blinding

Randomisation in an RCT consists of generating random sequences, assignment of subjects to the arms of the study according to the sequence, and maintenance of the assigned groups until the final outcome is measured.

Suppose there are two groups for a trial planned for 6 weeks of treatment period – group A receives active treatment and B receives placebo. Assume that we have completed the recruitment for the trial and we are in the third week of trial period.

What will happen if the clinician treating the patient comes to know which arm his patient belongs to? He might tend to over react to side effects seen in those who receive active medications and downplay side-effects in the placebo group.

What will happen if the patient comes to know which arm she/he belongs to? He/she might tend to report less improvement if they are taking a placebo, or overrate improvement if they receive the novel intervention.

What will happen if the assessor rating the patient comes to know which arm his patient belongs to? He might tend to overrate positive outcomes in active treatment group and underrate ill effects in the same group

This will defeat the whole purpose of randomization.

Hence, from the time of allocation, until the date of treatment completion and measurement, we need the group assignment to be kept secret. This is called **BLINDING.**

Different terms used

- ***Open label***: All parties are aware of the treatment being received after randomisation.
- ***Single blind***: Either the patient or clinician (usually the patient) remains unaware of the treatment assignment.
- ***Double-blind***: Both the patient and investigator are unaware of the allocated treatment.
- ***Triple blind***: The patient, the investigator and those who adjudicate the study outcomes (the assessor) are unaware of the allocated treatment.
- ***Unblinding***: The disclosure, planned or unintended, of the allocation of one, a group, or all of the participants.

A sequence/random list is generated

Generation

↓

Allocation to placebo/active arms according to the sequence

Assignment

↓

Treatment starts. Fidelity maintained till outcome is measured

Maintenance

↓

Measurement of final outcome

Allocation Concealment

Keeping the random sequence secret from investigator, clinician and patient until assignment is completed.

Blinding

Keeping the allocation secret from the investigator, clinician and/or patient until measurement is completed.

SPMM Memory Corner

Blinding refers to the secrecy that follows the group assignment until the completion of the study

Blinding may not be possible or ethical for some trials.

Allocation Concealment

Randomisation in an RCT consists of generating random sequences, assignment of subjects to the arms of the study according to the sequence, and maintenance of the assigned groups until the measurement of final outcome.

Suppose there are two groups for a trial – A for active treatment and P for placebo. Assume that we need 10 patients for the trial. A random sequence has been generated using a computer. This sequence reads as follows

APPPAAAPAP

Suitable subjects who give consent are allocated successively according to the above sequence.

What will happen if the clinician or the patient knows the above sequence? Consider that we are at the stage of recruiting 4th subject for the study.

APP*P*AAAPAP

The clinician has just seen a patient whom he thinks might be suitable for the trial. But he may think This patient seems to be doing poorly – I will rather allocate him to treatment group. I will wait till we get one more patient recruited before I send this one to the trial.

Similarly, the patients could also wait till they get their turn for proper allocation. This will defeat the whole purpose of randomization. ***Hence, from the time of sequence generation to the time of allocation, we need the sequence to be kept secret.*** This is called **ALLOCATION CONCEALMENT.**

How is this done?

- Firstly, the person who generates the allocation sequence should not be the same as the person who recruits/assigns subjects to the trial.
- If possible, use a centralised or remote randomisation service (either by telephone, fax, email, the internet, a coordinating centre or site pharmacy)
- If same personnel are involved in generating sequence and assigning the groups, then Serially Numbered, Opaque, Sealed, Envelopes containing the random sequence is employed. **(SNOSE)**

SPMM Memory Corner

Allocation concealment happens before the subjects start the treatment by protocol

It is always possible to conceal the allocation properly.

Intention To Treat Analysis

- When conducting an RCT, at the end of a trial, the number of studied individuals who remain in the trial may be very different compared to the initial number who gets randomised at the start. Previously, it was a common practice to report the results on the basis of the final count rather than the initial count.
- The intention-to-treat principle (ITT) holds that participants must be counted in the groups to which they were initially randomized, regardless of whether they were compliant with the allocated intervention. This includes participants who were withdrawn from active treatment for any reason. In extreme cases members who crossed over and received the treatment from the opposite group also get counted in their original group for the analysis.
- The rationale is that in an RCT one sets out to estimate the broader effects of allocating an intervention to a group, not the narrow effects in a subgroup of participants who strictly adhere to it. In real life, very few people adhere to treatment plans in a strict manner.
- The ITT strategy gives a conservative estimate of the treatment effect compared with what would be expected in an ideal world of full compliance. Hence, ITT tends to make RCT results more *generalisable and pragmatic*. ITT may not suit those who are investigating *biological effects* of a drug via RCT model – those who do not take the drug cannot have a biological effect. But for those studies that report on effectiveness, ITT is invaluable. ITT analysis will favour equality in equivalence trials due to dilution. ITT is also difficult to interpret if number of crossovers is high.
- Also included in ITT principle is the problem of attrition – lost for follow-up subjects/ drop outs. As the researcher may not have data on those who were lost to follow-up there are many methods used to circumvent this problem;
- ***Ways of dealing with missing data:***

1. **Test the differences**: Testing for differences between the lost tribe (e.g. drop-outs) and the available subjects group. Baseline characteristics if comparable then one can substantiate the loss to some extent without further adjustment – though not always ideal.

2. **Use available data - 1**: The **last observation carried forward (LOCF**): This is only useful for longitudinal observations where continuous scales were used. e.g., in a clinical trial of fluoxetine for depression observed for 6 months, the 3rd months HAMD/MADRS values can be carried forward to 4th, 5th and 6th months and used in the final analysis.

   i. Many psychiatric diseases have a natural course of remission – LOCF does not allow for this.

   ii. It is 'too good' for controls and 'too stringent' for the patient group.

3. **Use available data - 2: Worst case scenario analysis** Assume that all those who were lost to follow-up did not improve on the trial intervention. Hence, the data of 'treatment failure' subjects are used as a proxy. These extreme assumptions are usually carried out as part of sensitivity analyses.

4. **Imputation of data – 1**: Replace the missing values with the **mean** of the group

5. **Imputation of data – 2: Hot deck imputation**, involves finding a similar person among the subjects and using that person's score.

6. **Statistical methods – 1:** We can use **multiple regression** to predict the missing values, based on several other variables in the dataset

7. **Statistical methods – 2: Growth curve analysis**: In this approach a line is fitted through the existing data points for each subject individually. At least 2 follow-up visits are required for this type of analysis. From 2 data points, we can derive the parameters for a straight line; for 3 data points, we can fit a quadratic curve; and with 4 points, a cubic curve.

8. **Statistical methods – 3: Random effects modeling:** According to Streiner (Can J Psychiatry 2002;47: 68–75), "the term "random effects" derives from the fact that, when we usually fit a regression line to some data, we assume that the parameters (the intercept and slopes) are "fixed"; that is, the same for all of the subjects. In a random effects model, we allow them to differ for each person. This means that we do not expect that the curves for people who drop out will be the same as for people who complete the study".

9. **Statistical methods – 4:** Sophisticated and accurate methods, called **multiple imputation**, have been used – these approaches use maximum likelihood and Monte Carlo procedures. If missing data are imputed, it is recommended that some sensitivity analysis be performed to ensure that study conclusions are not misleading.

Alternatives to ITT:

Per-protocol (PP) analysis: According to this only patient who sufficiently complied with the trial's protocol are considered in the final analysis; the antithesis of intention to treat analysis. It is also called "on treatment" analysis. PP analysis is often carried out in biological explanatory RCTs and as the secondary analysis in effectiveness trials.

Treatment-received (TR) analysis: Another approach is to analyse subjects according to the actual treatment received – not actual allocation made. But the effect of a random assignment is compromised here due to contamination of trial by one group of subjects moving to the other group.

ITT analysis must be the norm unless there is overwhelming justification for a different analysis policy (e.g. high proportion of participants found not to have the disease etc.)

Advantages of ITT:

- Retains balance in prognostic factors arising from the original **random** treatment allocation
- Gives an **unbiased** estimate of treatment effect
- Admits non-compliance and protocol deviations, thus reflecting a **real clinical** situation

Limitations of ITT:

- Estimate of treatment effect is generally **conservative** because of dilution due to non-compliance
- In **equivalence trials** (attempting to prove that two treatments do not differ by more than a certain amount), this analysis will favour equality of treatments
- Interpretation becomes difficult if a **large proportion of participants cross over** to opposite treatment arms

Requirements for an ideal ITT analysis

- Full compliance with randomised treatment
- No missing responses
- Follow-up on all participants

N of 1 trial

When to conduct an N-of-1 trial?

- The clinician is uncertain whether a treatment that has not yet been started will work for a particular patient.
- The patient has started taking a medication, but neither patient nor clinician is confident that a treatment is really providing benefit.
- Neither the clinician nor the patient is confident of the optimal dose of a medication the patient is receiving or should receive.
- A patient has symptoms that both the clinician and the patient suspect—but are not certain—are caused by the adverse effects of the medications.
- The patient has so far insisted on taking a treatment that the clinician believes is useless or harmful, and although logically constructed arguments have not persuaded a patient, a negative result of an n-of-1 RCT might.

When is N-of-1 RCT feasible in a patient?

- Patient is eager to collaborate in designing and carrying out an n-of-1 RCT
- The treatment has rapid onset and termination of action; the underlying condition must not be modified by the treatment – only symptom control must be possible.
- An optimal duration of treatment is feasible
- Patient-important targets of treatment are amenable to measurement
- The criteria to end the n-of-1 RCT are identifiable
- There is a pharmacist who can help with blinding
- There are strategies in place for gathering and interpretation of the outcome

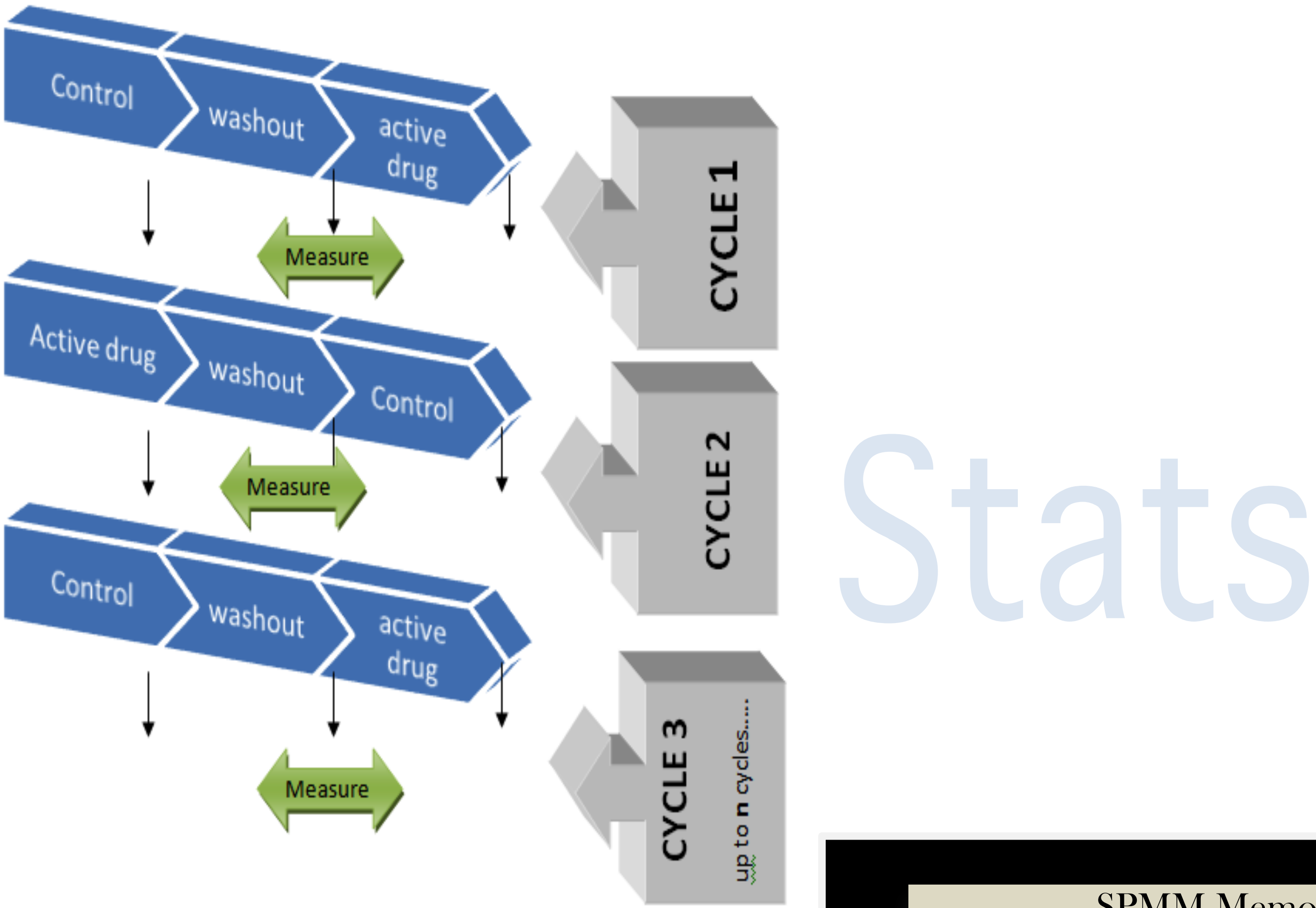
How to conduct an N-of-1 trial?

Both patient and doctor remain blind. The pharmacist delivers control and active drug in identical form. The allocation in each cycle is randomly determined preferably by a third party.

Regular measurement of the outcome is carried out. The patient and doctor agree when to stop the experiment. It is often done when a strong pattern has emerged with respect to the efficacy or lack of it. So there may be many cycles of treatment before it is ceased.

The data is then analysed to see which intervention fared better.

Guyatt G. et al. JAMA User's guide to medical literature, 2nd ed. McGrawHill Medicine 2007. Pg 180



SPMM Memory Corner

- N-of-1 trial has only one subject
- Generalisation of results to other patients is NOT possible
- Best EBM for answering therapy related question in a given patient, provided it is feasible

Causality & Confounding

Koch's postulates of causality

In 1890, Robert Koch formulated a set of postulates that should be met before concluding that a disease was caused by a particular 'germ' (bacterium). These postulates fit well for infectious diseases, but not generalizable to many non-infectious illnesses.

1. The agent must be shown to be present in every case of the disease by isolation in pure culture. (**Consistency**)
2. The agent must not be found in the case of another disease. (**Specificity**)
3. Once isolated, the agent must be capable of reproducing the disease in experimental animals. (**Biological coherence**)
4. The agent must be recovered from the experimental disease produced **(Predictive or experimental performance).**

Bradford Hill's criteria of causality:

To state A is a cause of B, we need to demonstrate all/most of the following assumptions.

1. **Temporal** association: A must precede B.
2. **Dose-response** association: higher the dose of A, higher must be the dose of B – and lower the dose of A, lower must be the dose of B.
3. **Specificity**: A must precede B and not precede C or D; similarly B must be preceded by A only and not P or Q. (this may not be always true as multiple causation is common in most chronic diseases).
4. **Consistency**: if A is present B must be present; if A is absent B must be absent.
5. A and B must have **plausible biological associations**.
6. A high **strength** of **association**.
7. An **absence** of **reverse causality** i.e. B must not be causing A.

Susser's criteria of causality:

Melvyn Susser grouped criteria to aid the judgment of causality into 5 categories (with many subcategories) and proposed 3 as essential.

The 5 categories are strength, specificity (both in the cause and in the effect); consistency (measured using replicability); predictive performance; and coherence or plausibility (with theoretical; factual; biological; statistical coherence as subcategories)

1. **Association**: Judged statistically by the strength of probabilities based on preset expectations of chance occurrences, and the consistency upon replication. The absence of association refutes causality.
2. **Direction (of prediction)**: The change in one state assumed to be the effect must be consequent on change in independent antecedent states presumed to be the cause. According to Susser, this directionality is the most critical and ultimately defining causal property.
3. **Time Order**: Reversal of time-order assures elimination of a putative cause

Rothman's Sufficient-Component Model:

1. In epidemiology, the entity that is termed as a cause is often not a single component, but a minimal set of conditions that inevitably produces the outcome. The term **sufficient cause** refers to this minimal set without which the outcome would not have occurred in a specific instance.

2. Each element that makes up a sufficient cause is called a **component cause***.* The outcome will not occur by that causal pathway if any one of the components is missing within a given sufficient cause model. This is an important idea in public health and prevention medicine, as it suggests that it is not necessary to identify all of the component causes in order to prevent a disease outcome.

3. There may be many sufficient causes for a given outcome. A component cause that must be present in every sufficient cause for the outcome to occur is called a **necessary cause**. For example, H. Pylori is necessary for a gastric ulcer to occur though on its own it may not be sufficient.

In Rothman's model, a sufficient cause is represented by a complete circle ("causal pie"), the segments of which represent component causes, one of the components being a necessary cause.

Web of Causation
In 1960, MacMahon and Pugh first employed the metaphor web of causation to describe the complexity of the relationship among antecedents when investigating the cause of a disease. In the causal web concept, there are mutual interactions among risk factors, which, therefore, have both direct and indirect effects on the outcomes. This model was further expanded by Murray & Lopez, who separated **proximal risk factors**, with established pathophysiological mechanisms from **distal risk factors** modelled on the basis of the empirical evidence. This model is also called hierarchical causal model.

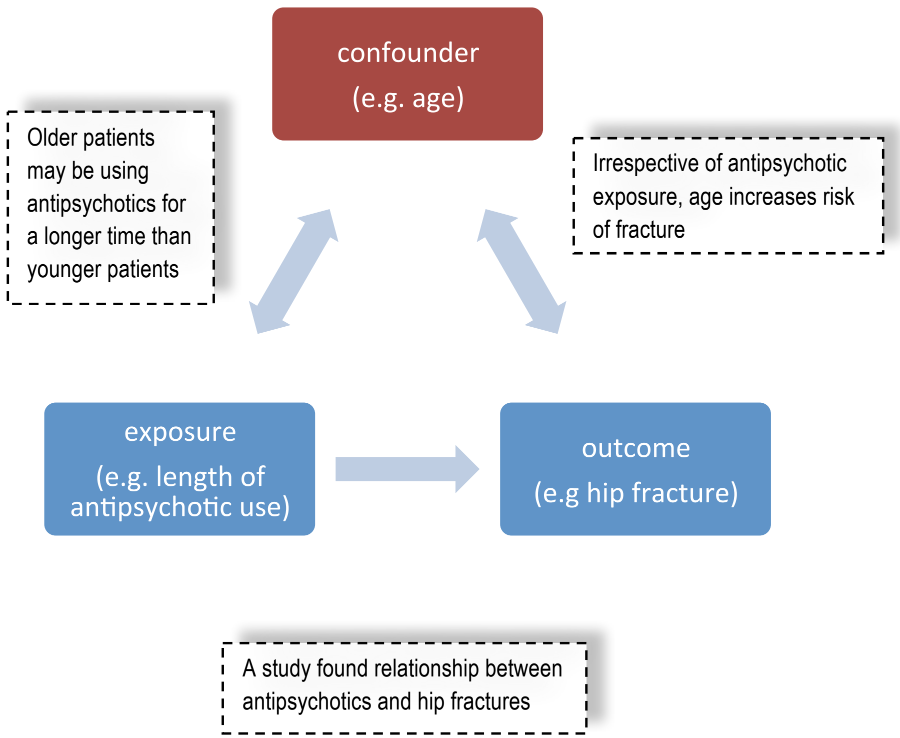
Explaining a study result:
A positive result from a study can be explained by any of the following factors:

1. Selection differences (bias) between two groups
2. Measurement differences (bias) between two groups
3. Confounding factor causing indirect association
4. Chance causing a spurious association (type 1 error)
5. Presence of a true causal association

Confounders:

A study is conducted to analyse the effect of a variable (independent variable) on an outcome of interest (dependent variable). Under ideal conditions, if the outcome is more often seen in a group with the independent variable than the one without it, then one can presume that there is an association between the outcome and the study variable.

But sometimes, the above result may not be due to the influence of the independent variable. Instead, it may be due a third factor called confounder.



For a characteristic to be a confounder in a particular study, it must meet four criteria.

- Δ It must be related to the exposure in some way
- Δ It must be related to the outcome in terms of prognosis or susceptibility.
- Δ It will not be on the causal pathway between exposure and outcome i.e. antipsychotics do not cause fractures by increasing one's age!
- Δ The distribution of the characteristic must be different in the groups being compared. The difference can be in terms of either the central value (e.g. mean) or the variability (e.g. Std deviation) of that characteristic.

A well-conducted analytical study must have the following elements in order to withstand a confounding effect:

1. There must be a systematic effort to identify and measure potential confounders
2. The data on the distribution of potential confounders across the groups must be available. Group differences must be)

Confounders cannot be eliminated – but controlled for reduced.

How to control confounding?

Method	*In the design of study*
Restriction	Avoid including the group with significant confounding influence in the study: e.g. do not include any older people in the hip fracture study. This will limit sample size.
Matching	Make sure that the confounders are equally distributed: e.g. for each older patient with high antipsychotic use, recruit one younger patient with heavy exposure in the hip fracture study
Randomisation	This will ensure the two groups are more likely to be similar in terms of confounder distribution. But one must check if this is the case after recruitment If they are not consider stratification/ multivariates. Unlike other methods listed, randomisation can help control both known and unknown confounders.

Source: ***Elwood, M. Critical appraisal of Epidemiological studies and Clinical Trials***

Method	*In the analysis of the study*
Stratification	When analyzing results, tabulate data for various levels/ categories of exposure to confounder separately and analyse the variable influence; e.g. analyse the association of antipsychotics and hip fractures in three different age groups individually. Thus, subgroups are created for analysis
Multivariate methods	This includes regression methods to analyse the effect of various confounders. This produces adjusted rates and crude rates.

Source: *Elwood, M. Critical appraisal of Epidemiological studies and Clinical Trials*

Effect modifier is a third factor, existing as a sub-group often but not related to outcome or exposure. This can be eliminated by stratified analysis.

- A factor, E, is said to be an effect modifier of a relationship between a risk factor, A, and outcome B, if the strength of the relationship between the risk factor, A, and the outcome B varies according to the levels or values of E.
- If E is an effect modifier, then it is important to report the strength of the A-B relationship for specific values of E. If the strength of the A-B relationship does not vary greatly among the levels of E, it may not be an important effect modifier.

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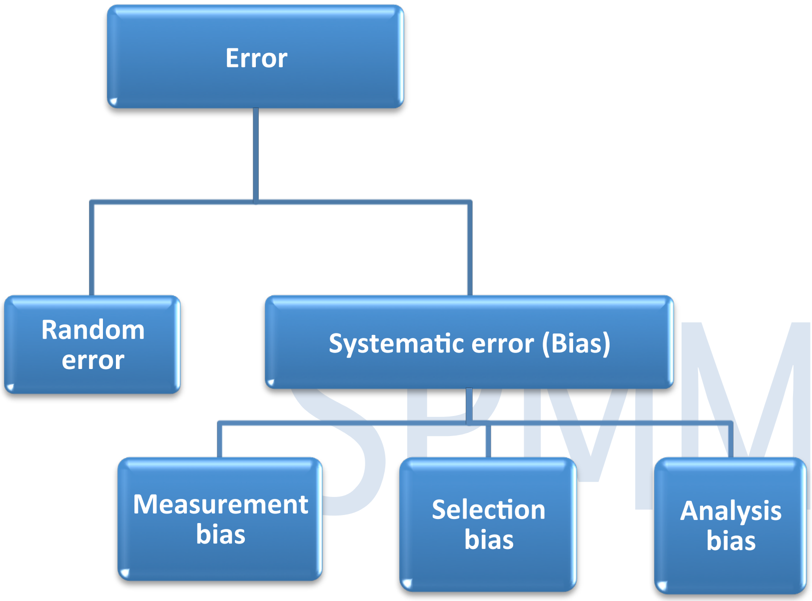
SPMM Memory Corner

Confounders affect internal validity of a study – so they result in bias

Known confounders can be controlled / accounted for when designing a study / analysing data. Unknown confounders can still influence the results of a well-conducted study

Bias: Introduction



Is the study valid? This is the first question which a person asks after reading a paper.

Firstly, if the result of a study remains close to the truth, then the study is said to have **internal validity**.

Secondly, if the results of the study are applicable to the clinician seeking the evidence, then the study is said to have **external validity**– or generalisability.

Validity of a study may be compromised due to three reasons

1. **Random error (chance/imprecision)**
2. **Systematic error (bias)**
3. **Confounding**

A ***random error*** is due to the play of chance. i.e. our finding can be inaccurate, just by chance. All studies are prone to random error because they use samples drawn from a population to *estimate* what is occurring in the population. The probability of this is assessed using statistical measures such as p-values and confidence intervals. Random error can be reduced by repeating/ replicating the study elsewhere.

A ***systematic error*** or ***"bias"*** is an error in the way we select our patients, measure our outcomes or analyse our data. Bias may result in inaccurate results. This type of error is predictable and does not reduce by itself by mere repetition of the study elsewhere.

SPMM Memory Corner

- Bias risks researchers drawing conclusions that are false.
- Bias can lead to an over or under-estimation of an effect.
- Bias may be introduced by poor study design or poor data collection.
- Bias cannot be "controlled for" at the analysis stage.

Types of bias

Bias can be classified by the direction of the change they produce in the study outcome.

Toward the null bias or **negative bias** yields estimates closer to the null value (i.e. no difference between two groups), whereas away from the null bias (**positive bias**) produces the opposite, higher magnitude of estimates than the true ones. An exaggeration of these biases can induce a switch-over bias, or change of the direction of association (for example, a true OR >1 becomes <1).

Bias can creep into a study during the various stages

1. Selection of participants – **Selection bias**
2. Measurement of variables – **Measurement bias**
3. Analysis of data – **Analysis bias**

Selection bias is said to occur when the selected groups in a study differ in important factors other than the study variables. For example, when comparing an antidepressant drug with placebo in a trial, if patients assigned to one group are older, or more chronically depressed, there is a selection bias. Here the study population does not represent the target population. This type of sampling bias is minimised by adequate randomisation. Uneven diagnostic procedures in the target population could also produce selection bias.

Other types of selection bias include

a. **Berkson bias** or the admission rate bias results from a difference in the rates of admission of cases and controls due to the influence of the exposure. E.g. In a case-control study of smoking and dementia, the association will tend to be weaker (or even absent) if controls are selected from a hospital population (because smoking causes many diseases resulting in hospitalization) rather than from the community.
b. **Neyman bias** is an incidence-prevalence bias. While ascertaining causation, one must look for an association between a risk factor and incidence – not prevalence. If a case-control study evaluates a risk factor that makes a person die quickly, then this will be underrepresented if 'prevalent cases' are studied instead of 'incident cases'.
c. **Response bias** exists when persons who respond to an invitation to participate in a study differ systematically from those who do not respond. The 'healthy volunteers' are often healthier than the general population. This is particularly relevant when evaluating screening tests.
d. **Unmasking bias** is said to exist when a risk factor unmasks (rather than causes) an event which it is suspected to cause.

e. **Lead-time bias:** Lead-time is defined as the difference in time between the date of diagnosis with screening and the date of diagnosis without screening. Unless lead time is accounted for, survival time should not be compared to an unscreened control group of patients. Otherwise, the increase in survival time due solely to the advanced date of diagnosis will result in lead-time bias.

a. **Referral bias** refers to the variation in concentration of rare exposures or diseases between patients in primary and secondary care.

b. **Diagnostic purity bias** refers to the exclusion of comorbidities resulting in a non-representative sample, especially problematic in ***RCTs***.

c. **Membership bias** refers to case identification using members of patient's organizations (e.g. a local self-help group) leading to systematic differences and non-representativeness.

Measurement Bias results when data are not collected in a uniform fashion. For e.g. when cases are interviewed in person while controls are interviewed over the telephone. This type of bias is minimised by ***"blinding".***

Types of measurement bias include

a. **Recall bias**: Subjects often recall risk factors differently depending on their disease status. Case-control studies are particularly vulnerable to this type of bias.

b. **Reporting bias** results when a larger percentage of either case or control subjects are reluctant to report an exposure due to attitudes, perceptions or other concerns

c. **Observer bias** can occur whenever a researcher either knowingly or unknowingly evaluates a variable depending on the status of the individual under study. For e.g. when the research observer knows that a subject is on placebo, he may rate him higher on depression in a trial.

d. **Surveillance bias:** Disease may be better ascertained in a monitored population than in the general population

e. **Work up bias (verification bias):** During assessment of validity of a diagnostic test, the execution of the gold standard test may be influenced by the results of the assessed new instrument; e.g. the reference test may be less frequently performed when the test result is negative.

f. **Misclassification bias:** In extreme cases measurement bias may lead to misclassification. Cases may be misclassified as controls or 'exposed group' may be misclassified as 'non-exposed'. Such misclassification amounts to bias only if it is differential i.e. one-sided. Errors in measurement instruments may lead to non-differential misclassification (both sides are affected equally), which often leads to a reduction in the observed magnitude of association rather than producing biased results.

g. **Desirability bias** – patients may choose socially desirable answers to provide during data collection, distorting the true picture. This leads to reporting bias.

h. **Hawthorne effect** refers to observed respondents minimizing perceived deviation from the norm. Occurs especially in ***cross-sectional surveys*** using questionnaires.

Analysis Bias is said to occur when participants change group (*contamination bias*) or are lost to follow-up (*attrition bias*) during the study. This can be minimised by doing ***"intention to treat"*** analysis.

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Page L, Henderson M *Appraising the evidence - what is selection bias? Evidence-Based Mental Health* 2007;**10**:67-68

Qualitative Studies

- Not all data can be presented as numbers. For example people's attitudes, values and wishes cannot be quantified and analysed without loss of significantly important elements. In these cases, we use qualitative research.
- Qualitative research largely developed from social research. The important questions that can be studied qualitatively are;
  - Why people behave the way they do
  - How opinions and attitudes are formed
  - How people are affected by the events that go on around them
  - How and why cultures have developed in the way they have
  - The differences between social groups

(Retrieved from http://libweb.surrey.ac.uk/library/skills/)

Types of qualitative research (QR):

Five major types of qualitative research design are outlined. They are:

1) **Phenomenology:**

This type of QR studies many phenomena that happen around us all the time but go undefined and not characterised clearly. For example, we know that lots of people are carers. But what does "caring" actually mean and what is it like to be a carer? This can be answered by constructing a phenomenology study in QR.

(N.B.: Phenomenology of QR not to be confused with descriptive psychopathology and Jaspers!)

2) **Ethnography**

The term refers to "portrait of the people" and it is a methodology for descriptive studies of cultures and peoples who have something in common such as geography, religion or any shared health experience.

Emic perspective (emic view): Used by ethnographers to refer to the perspective of an individual from a specific cultural group about his own group.

Etic perspective (etic view): Refers to the perspective of an individual outside a specific cultural group about the studied group.

3) **Grounded theory:**

(Defined by Glaser & Strauss)

Grounded theory

This term is used in two ways:

1. To describe the qualitative analysis in general where theories generated are 'grounded' in the collected data.
2. To describe a precise technique of generating theories/categories.
   i. Initially, a constant comparison of the collected data takes place in different stages of the data collection.
   ii. From these comparisons, categories are generated until nothing new is revealed, i.e. the point of theoretical saturation is reached.
   iii. Later these categories are tested by collecting further fresh data on the basis of the new categories made – this is called theoretical sampling.

A qualitative research paper must state this methodology explicitly.

One example of grounded theory is a theory about the process of grief. Researchers observed that people who were bereaved progressed through a series of stages such as denial, anger, bargaining, and depression and at last acceptance (**DABDA**). This is not a new phenomenon that happened in human grieving response, but the research formally acknowledged and described the experience. Now we use the new knowledge derived from grounded theory, to understand the experience of bereavement in a fresh patient coming to our clinic to help her come to terms with her loss.

4) **Case study:**

Case study research is one of those research approaches which fall in between qualitative and quantitative approach. If qualitative methods such as in-depth interviews, are employed, a case study is best described as a qualitative research. (see below). A 'case' in a case study approach could be single unit e.g., a person, an organisation or an institution. Some research studies describe a series of cases.

5) **Participant action research:**

PAR (or simply action research) involves individuals and groups researching their own personal beings, socio-cultural settings and experiences. The clear cut demarcation between "researcher" and "researched" that is found in other types of research may not be so apparent in action research. For example, a PAR study may involve researchers working with carers of schizophrenia patients to identify which family or environmental situations provide substantial challenges to the service users.

- Some situations where qualitative research is useful are,
  1. When exploring the nature of the problem to be studied; i.e. before planning quantitative research
  2. To generate hypothesis based on observations made.
  3. Investigating anomalies or oddities observed during clinical practice
  4. Examining policy implementation
  5. Collating service users/ carers/ staff views

Data collection methods in qualitative research: Three basic techniques are employed:

1. Interviews. 2. Observation. 3. Document analysis.

Interviews

- Interviews can be highly structured, semi-structured or unstructured.
- In **structured interviews,** a tight schedule of questions may be used with the intention to elicit one of the few limited range of responses, similar to a questionnaire.
- **Semi-structured interviews** (also called focused interviews) involve a series of open-ended questions based on previously selected topic areas the researcher wants to cover. Though there is a schedule of topics to cover, the exact way questions are phrased is not premeditated or dictated by a list.
- **Unstructured interviews** are also referred to as "depth" or "in depth" interviews where the interviewer has some limited number of topics in his mind but does not follow any order and frames the questions on the basis of the interviewee's previous response.
- The interviewees may be individuals or groups, according to the need of the researcher.
  - ***Focus groups:*** Conducting interviews in a group format with some predetermined structure for such groups, in order to provide a focus for discussion, is called 'focus groups' (Kitzinger, 1995). Focus groups usually contain a small number of members selected giving views on the chosen topic; the discussion here is semi-structured guided by key points. These are useful in gathering opinions of service users and providers. The following are features of focus groups:
    1. useful when limited resources prevent individual interviews
    2. used when the studied individuals share a common factor where collecting views of several of them may be useful; these factors should be important to the topic of investigation
    3. If group interaction may allow greater insights to be developed, focus groups may be used
    4. members must usually be 6 – 10;
    5. not a good idea to have just one group for the whole research – many small groups must be included in the research

Observation

- At times, qualitative data collection approaches are not complete by direct interaction with people. For example, in interviews participants may be asked about how they behave in certain situations but the reported behaviour cannot be guaranteed in real life situations. So, observing them in the real life situations may be more reliable.
- During observation, data can be collected using written descriptions, video recording or photographs and artefacts.
- **Field notes** refer to what the researcher writes down when collecting data on the field. This is later reviewed out of the field and collated.
- The observer may either be a complete outsider or participate in the observed activity themselves, to closely mimic real life situations if needed.

Data from documents:

Documents used for this purpose may be either informal documents such as diaries, letters or formal documents such as case notes.

Analysing the data collected in qualitative research:

- Interviews → transcribed to texts → coding categories generated.
- Data collection process in qualitative research includes **transcribing**. Transcribing is the procedure for producing a written version of the interview. It is a full "script" of the interview.
- Transcribing is a time-consuming process. The estimated ratio of time required for transcribing interviews is about 5:1 (i.e. a 10 minutes interview may need 50 minutes to be transcribed).
- **Coding:** This refers to the process of transforming raw data into a standard format to enable data analysis. Coding involves identifying recurrent words, concepts or themes. It may involve attaching numerical values to categories.
- Various levels of data analysis are used.

⊙**Content analysis:**

  - Content analysis aids in the categorisation of spoken, written or observational data, in order to classify, tabulate and summarise it.
  - The content can be analysed on two levels. The basic descriptive account of the data: this is what was actually said with nothing interpreted or assumed about it. (Also called **manifest level** of analysis).
  - The higher level of analysis is often concerned with what was meant by the response, what was inferred or implied. (Also called the **latent level** of analysis).
  - This can be done via enumerating procedures such as counting word frequencies, analysing annotations, intonations, etc.

⊙**Conversation analysis:**

  - Here combined audio and visual recordings are used to study interactional practices in particular settings (contextual analysis takes place).

⊙**Discourse analysis:**

  - This centres on the interpretation of social discourse in the context of the occurrence via analysis of personal accounts or written autobiographical documents.

Methods of assessing validity of qualitative research:

Triangulation : Triangulation compares the results from either two or more methods of data collection or data sources. The researcher looks for patterns of convergence to develop or corroborate an overall interpretation. In investigator triangulation, more than one investigator is involved, and findings are cross-validated.

Respondent validation: ("member checking") This includes techniques in which the investigator's account is compared with those of the research subjects to establish the level of correspondence between the two sets.

Reflexivity means sensitivity to the ways in which the researcher and the research process have shaped the collected data, including the role of prior assumptions and experience, which can influence the results.

Deviant case analysis refers to paying attention to, searching for, and discussing elements in the data that contradict, or seem to contradict, the emerging explanation of the phenomena under study.

(Excerpts retrieved from Mays, N., & Pope, C. (2000). Qualitative research in health care: Assessing quality in qualitative research. *BMJ: British Medical Journal*, *320*(7226), 50)

	QUALITATIVE RESEARCH	QUANTITATIVE RESEARCH
Type of questions answered	Why? How? In what way?	How much? How many? How often? To what extent?
Core elements of data collection	Opinions, experiences and feelings of individuals	Frequency, distribution and causation of diseases, etc. and effect of interventions
Nature of data	Subjective data; a repeat attempt to simulate the methods may/ may not produce same results	Objective data that can be collected again if needed using same methods
Relationship to phenomenon studied	Describes social phenomena as they occur naturally	Describes effects of interventions on naturally occurring phenomenon or nodal points where natural phenomenon could be modified e.g. diagnosis
Sample	Purposeful sampling; no predetermined sample size – sampling takes place till theoretical saturation occurs	Representative – hence random or matched samples preferred with a predetermined numbers according to power of the study
Approach used	Inductive approach – applying general facts to sub-samples to understand an experience	Deductive approach – making generalisations from samples.
Time required	Data collection is time-consuming	Data collection is comparatively (only comparatively!) less time-consuming
Sample size feasibility	Small samples	Larger samples can be employed if needed based on resource availability
Representativeness	The sample is interested in outliers – seeks to include them to understand their experiences too.	Sampling seeks to demonstrate representativeness, so outliers are not preferred generally.
Purpose served	Phenomenological i.e. 'describing.'	Scientific i.e. 'explaining.'
Consideration of multiple factors	Holistic	Reductionist; but may seek to explore many such 'reduced' individual factors to build a picture of the sum.
Hypothesis	Research may generate new hypothesis	Research tests a proposed hypothesis.

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Secondary Research

Secondary research refers to research conducted using data from other studies. When conducting secondary research, individual patients are not recruited but eligible studies are 'recruited'. Two important types of secondary research are 1] Narrative reviews 2] Systematic reviews

Narrative reviews:

These reviews are usually carried out by experts in the field of study and are more or less aligned with the expert's own opinion on the issue supported by selected research evidence. These reviews are generally broad without a narrow or specific clinical question and are subjective. They carry a higher risk of being prone to bias than systematic reviews. Literature is not searched methodically but rather in a random manner to suit one's flow of argument. Because of the above, narrative reviews cannot be replicated (unless repeated by the same author!). Nevertheless narrative reviews can provide a good introduction to a topic, stimulate interest and controversies and can even collate existing data for generating new hypotheses.

Systematic reviews:

Systematic reviews follow rigorous steps in identifying relevant literature, appraising individual studies and analysing suitable data to synthesise a conclusion. The characteristics of a systematic review are

1. Study a focused narrow question
2. Comprehensive & specific data collection (see table)
3. Uniform criteria for study selection (see table)
4. Quantitative synthesis of data (may or may not be possible)

The 4th feature is optional as not all data are 'combinable' to produce a final common result. Also in some situations good quality data may not be available to combine and synthesise a concluding result. If the statistical combination is possible, then such procedure of quantitative synthesis of individual study data is called a **meta-analysis.**

	Narrative review	Systematic Review
Question	Broad in scope; general and vague sometimes	Focused and narrow
Search methods	Not usually specified; comes from an experts familiarity with the literature	Comprehensive & explicit; using well defined search strategies
Appraisal of individual studies	Variable; usually sails on the expert's opinion	Rigorous appraisal is made to each individual study included irrespective of the direction of the outcome
Synthesis of results	Often provides only a qualitative summary of studies	Quantitative summary (using meta-analysis) is possible
Resources	Comparatively less time consuming; but requires an expert in the field to guide the process	Highly time consuming; no need for field expertise; in fact non experts can carry out an unbiased search than the experts at times.

Individual trials' quality: If the primary studies included have poor quality, then even if sound methodological and statistical procedures are followed the outcome may be meaningless. This is called GIGO principle of ***'garbage in, garbage out'***!!

The quality of individual trials can be assessed by considering

1. The nature of the patient sample,
2. Outcomes studied,
3. Length of follow up and
4. Comparability (e.g. dosage) of treatment and
5. Methodological factors such as
    1. Adequacy of sample randomisation, 2.adequate concealment and control of intervention, 3. Analysis with the intention to treat principles, and 4. Having an objective, blinded, outcome assessment

A PROPER LITERATURE SEARCH MUST HAVE FOLLOWING CHARACTERISTICS:
1. Using multiple databases – no just limiting to MEDLINE only or CINAHL only. 2. Cross checking the references list of each individual study retrieved by direct search 3. Hand searching for materials unidentified online. 4. Approaching experts to comment on any missing studies. 5. Identifying grey literature – i.e. the unpublished literature such as conference abstracts, presentations and posters.

The inclusion criteria for studies in a systematic review should consider
(1) Types of study designs to include (2) Types of subjects to include (3) Types of publications (4) Language restrictions, if any (5) Types of interventions (6) Time frame for included studies

Meta-analysis:

- A meta-analysis is most often used to assess the clinical effectiveness of healthcare interventions.
- Hence RCTs are the most frequently combined studies for a meta-analysis though it is possible to combine other study designs.
- A meta-analysis may either use summary data or individual patient data. For example, if one wants to do a meta-analysis of 3 studies each having 50 patients enrolled, using summary method your N will be 3. In case if one is following individual patient data method this requires contacting the authors of each of the 3 trials and requesting the data of the subjects. Though this is ideal, the inability of investigators to supply data, and the increased costs associated with the retrieval of individual patient data make researchers carry out summary data method more often.
- The four basic steps for conducting a meta-analysis include
  (1) Literature search,
  (2) Establishing criteria for including and excluding studies
  (3) Recording of data from the individual studies, and
  (4) Statistical analysis of the data.

Quality of a meta-analysis or systematic review depends on the following factors
•A comprehensive literature search (see below) •Clearly defined eligibility criteria for the studies to be included (see below) •Clearly defined & strictly adhered protocol. •Predetermined criteria related to the quality of trials •Assessing the quality of a study can be difficult since the information reported is often inadequate for this purpose. In these cases, a thorough sensitivity analysis must be conducted. •Discussion and analysis of statistical and clinical heterogeneity.

Combining individual trial data:

- Methods used for meta-analysis use a weighted average of the results.
- **Weighting** refers to the significance attached to each study based on sample size, precision, external validity (the extent to which results are generalisable) and methodological quality. It is assigned independently of the outcome of a study.
- Two models of analysis:
  - When combining the outcomes from different studies, one may use a fixed and/or random effects model.

Fixed effect statistics

- Mantel-Haenszel and Peto ratios are used in fixed effect analysis
- MH: useful even when wide differences exist between individual studies in ratios of the size of two groups:
  - hence useful for review of cohort / case-control designs too
- Peto – mostly restricted to reviewing RCTs as it can produce biase results in unequal groups

- A **fixed effects model** assumes that all the studies share the same common treatment effect (homogeneous) while a **random effects model** assumes that they do not share the same common treatment effect.
- In fixed effect analysis
  - The inference is restricted to included set of studies.
  - It assumes that only random error within studies could explain observed differences.
  - It ignores between-study variations (hence heterogeneity).
  - So this can be applied only if heterogeneity can be safely excluded by testing for it.
- Random effects analysis
  - It assumes that each study shows a different effect which are normally distributed around true mean.
  - This assumption gives proportionally greater weight to smaller studies.
  - Hence, this model is susceptible to publication bias and results in wider less precise confidence intervals.
- In the absence of significant heterogeneity (often calculated using the **Q statistic**), both the fixed and random effects models yield similar confidence intervals.
- Although neither of two models is wrong or inaccurate, a significant difference in the combined effect calculated by the two models will be seen only if studies are very heterogeneous.
- In the presence of significant heterogeneity, the random effects model will yield wider confidence intervals which may be more representative.

Clinical heterogeneity

Heterogeneity can occur for clinical reasons in trials. For example, even if the same outcome / end result is considered (e.g. increase in insight), the interventions may be diverse, such as providing psychoeducation sessions, leaflets, videotapes, audiocassettes or talking to the family. Though the results can be combined theoretically (all odds risk differences having weighted mean) this would not be meaningful. The interventions are so different that combining them does not make clinical sense. This is an example of clinical heterogeneity. Other circumstances that may give rise to clinical heterogeneity include differences in selection of patients, the severity of disease and dose or duration of treatment. Clinical heterogeneity is describable but not measurable. (Retrieved from http://www.bmj.com/content/334/7584/94)

Methodological Heterogeneity:

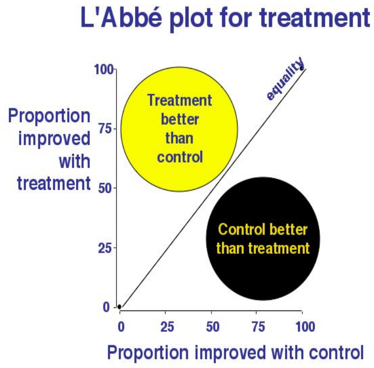
This refers to heterogeneity resulting from the differential use of study methodology. These may lead to different conclusions in different studies, even though the clinical characteristics are the same. Methodological heterogeneity is describable but does not need quantification.

Statistical heterogeneity

In a systematic review, individual studies may purport to measure the same outcome but often report results that are not consistent with each other. Some trials may show a benefit while others show a lack of response, or the trials may be inconsistent in the size of the observed difference between the placebo and active treatment arms. This variation is called statistical heterogeneity. A homogenous sample refers to a set of individual studies that have comparable outcomes without much variation (i.e they do not disagree greatly among each other). Heterogeneity refers to the presence of significant variation among the individual studies in a sample.

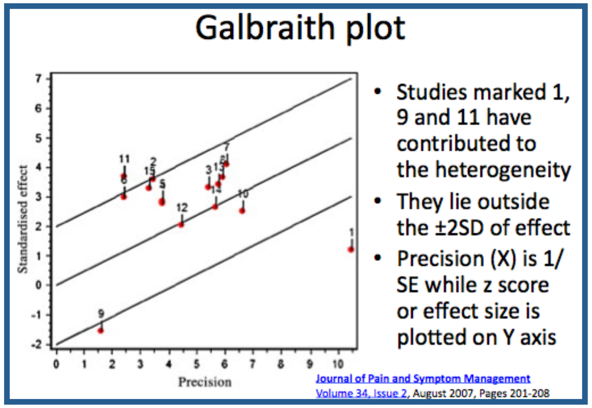
Test for statistical heterogeneity across studies:

- Heterogeneity may be judged graphically or be measured statistically.
- Forest Plot & L'Abbé plot can be used to explore the inconsistency of studies visually. **Forest plot** is described elsewhere (see blobbogram).
- **L'Abbe plot is nothing but a modified scatter plot wherein** CER is plotted aganst EER from individual trials included in the meta-analysis. Trials that report experimental treatment to be superior to the control arm (EER > CER) will be in the upper left of the plot, between the y-axis and the diagonal line of equality. If experimental is no better than control then the point will fall on the line of equality (EER = CER), and if control is better than experimental then the point will be in the lower right of the plot, between the x-axis and the line of equality (EER < CER). This provides a visually clear idea of how heterogeneous the results are.



→L'Abbe plot:

Note that proportions improved in both arms of individual studies are plotted across the X and Y axis. ***(From Bandolier website)***



- The **Galbraith plot** is an alternative to a forest plot as where on the horizontal axis one plots 1/standard error (1/SE = a measure of **precision**) of the study effect estimate while on the vertical axis one plots the study effect estimate divided by its standard error (e.g. log odds ratio/ SE = called **standard normal deviate**).
- Statistical procedures can test whether the results of a study reflect a single underlying effect (homogenous) or a distribution of effects (heterogeneous). A major limitation of this approach is that the statistical tests lack the power to detect heterogeneity in most meta-analyses.
- **Chi-square** tests of heterogeneity can be either Q test or I^2 test.
- The chi-square tests provide a test of significance for heterogeneity, but do not measure it.
- As a rule of thumb (Greenhalgh, 2001) a χ^2 value on average is equal to its degree of freedom, which is n-1 (number of studies – 1). So for a meta-analysis of 8 trials, n-1 is 7, and χ^2 value of less than or = 7 means no significant heterogeneity is present. But because of the low power of the test, one cannot say that homogeneity is present either!
- A well-known approach to quantify heterogeneity is **Cochran's Q**, calculated as the weighted sum of squared differences between individual study effects and the pooled effect across studies.
- The **I^2 statistic** describes the percentage of variation across studies that are due to heterogeneity rather than chance.
- In these chi-square tests, high P suggests that the heterogeneity is insignificant and that one can perform a meta-analysis.
- It is important that heterogeneity should not be considered as a mere problem to carry out meta-analysis; it must be investigated, and reasons must be dissected (similar to transference analysis in psychodynamics!).
- In case there is a significant heterogeneity, one can either study the reasons such heterogeneity, which can be
    - 1. Clinical differences,
    - 2. Methodological issues such as problems with randomization,
    - 3. Early termination of trials,
    - 4. Use of absolute rather than relative measures of risk, etc. and

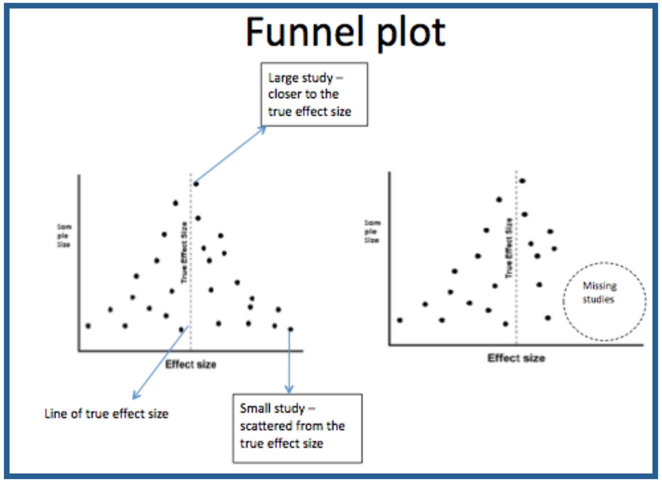
- 5. Publication bias.

Or, one can resort to random effects analysis. Procedures such as meta-regression analyses and subgroup analysis can throw light on the causes of heterogeneity (see below).

Publication bias:

- This refers to the tendency for authors to submit and/or editors to publish only those studies that yield statistically significant results (i.e. ‘positive’ and rejecting the null hypothesis).
- There are several methods available to diagnose the presence of publication bias in the literature. Most of these methods operate on the basis that small studies are more susceptible to publication bias and may, consequently, show larger treatment effects.
- Application of tests for publication bias in the presence of heterogeneity (described below) is not valid, and may lead to false-positive claims for publication bias (Ioannidis, 2007).
- When all available studies are equally large (i.e., have similar precision), the tests are not meaningful.

__Funnel plot:__ The most common method to detect publication bias is to investigate the presence of asymmetry in (inverted) funnel plots. A funnel plot shows the relation between the effect size and precision of the individual studies in a meta-anaysis. Small studies are more likely to remain unpublished if their results are nonsignificant or unfavourable, and larger studies get published regardless prducing an asymmetrical funnel-plot. Although visual inspection of funnel plots is often unreliable, several statistical tests can be used to quantify the asymmetry.



__Failsafe N:__ Meta-analytic results could be wrong, due to non-significant studies that remained in researchers’ ‘file drawers’ and not published. One can speculate whether the inclusion of these studies would nullify the statistical significance of an observed mean effect in a meta-analysis. Rosenthal & Cooper developed a formula to calculate the number of zero-effect studies (no effect) that would be required to nullify the mean effect seen in a meta-analysis. This number was termed the failsafe N or file-drawer number.

__Other tests for publication bias__: Either linear regression method or correlation method (Tau) can be used to quantify the publication bias. Details of such methods are beyond the scope of MRCPsych exams.

__Cumulative meta-analysis:__ This can be used to assess the potential impact of publication bias in tilting or nullifying an effect. A cumulative meta-analysis is done in the following manner:

- The studies are sorted from largest to smallest (or from most to least precise).
- A meta-analysis is initially run with one study, then repeated with a second study added, then a third, and so on;
- A forest plot/blobbogram (see elsewhere in the notes) is plotted as one goes along the process of accumulation. Here the first row will show the effect based on one study, the second row shows the cumulative effect based on two studies and so on.
- If the point estimate stabilizes based on the initially added larger studies, then there is no evidence that the smaller studies are producing a biased overall effect.
- If, however, the point estimate shifts when the smaller studies are added, this can be considered as an evidence of bias and one can also see the direction of the biased effect.

→Funnel plot:

Note that the funnel is well formed in this example. X axis of funnel plot carries a measure of effect size (OR, log OR, Cohen’s d etc). Y axis carries a measure of precision such as sample size, inverse of standard error (1/SE) etc. ***(From Cochrane database training section)***

The trim-and-fill procedure: This was developed by Duval and Tweedie (2000). This helps to assess whether the effect would change if the bias were to be removed. Trim and fill procedures is a kind of sensitivity analysis where missing studies are imputed and added to the analysis, and then effect size is recomputed.

(Other biases that can occur include **location bias** – studies not being located due to citation habits, the database used, key terms used or multiple replications of same data.

Inclusion bias refers to the reviewers' tendency to include studies they agree with.)

Sensitivity analysis: This refers to the analysis of the extent to which an outcome derived from research will change when 'hypothetical data' are introduced. One always wonders if an established result could be different if certain aspects of a research process were different. Such an exploration of 'what ifs' is called sensitivity analysis (Greenhalgh, 2001).This applies to mainly to economic analysis and meta-analysis. In a meta-analysis, there are various nodal points where such hypothetical data can be introduced. For example

1. What if there was significant heterogeneity (or absence of it) – how will this affect the outcome? One can do analysis in both random and fixed effects models.
2. Factors determining methodological quality e.g. numbers allocated in either arm, type of outcome measure used, case definition used, etc. can be manipulated for sensitivity analyses.
3. Publication bias can be examined using sensitivity analysis methods e.g. failsafe N.

Blobbograms:

- A blobbogram or **forest plot** presents the effect (point estimate) from each individual study as a blob or square (the measured effect), with a horizontal line (usually the 95% confidence interval, indicating the precision) across the blob.
- The size of the blob or square reflects the amount of information in that individual study (often the sample size);
- The length of the horizontal line represents the degee of uncertainty of the estimated treatment effect for that study.
- A mid-vertical line is often constructed to represent null effect for the differences in outcome; on either side lies the area favouring intervention A or B. When the blob or square lies on one side of this vertical line, this means that the outcome of that specific favoured the intervention marked on that side. If an individual study result is statisticsally significant then the horizontal lines representing uncertaininty will lie entirely on the same side of the blob, not touching or cross the midvertical line.
- A final synthesis of individual studies is represented in the form of a diamond lozenge. The position of the lozenge depends upon whether the intervention rates favourably or not at the end of the meta-analysis.
- The following figure is a forest plot

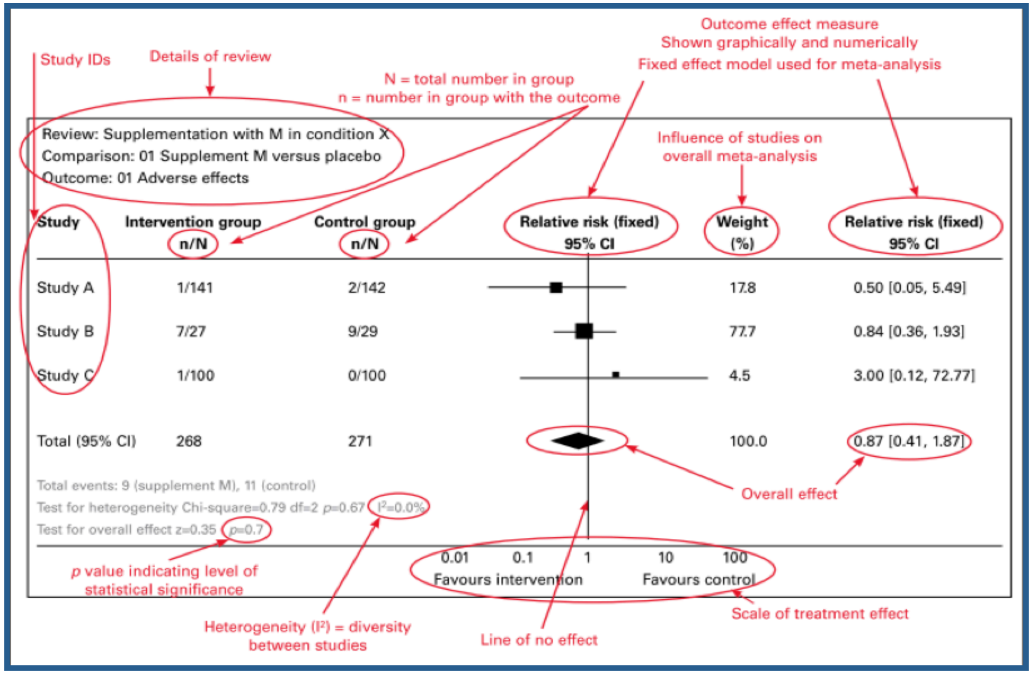
Forest plot: Things to note in a forest plot

1. The combined /pooled effect size (given by various summary statistics including OR, RR, standardised mean differences, Cohen's d etc)
2. Confidence interval of individual point (line width) estimates and combined estimate (lozenge width)
3. Weighting assigned to studies (rectangle size)
4. Heterogeneity of results (absence of heterogeneity indicated by vertical linearity of rectangles or non significant chi square test for heterogeneity)

American Journal of Psychiatry 2003; 160:1919–1928

Effect size:

- *Effect size measures for dichotomous variables:* In meta-analyses, relative risk and odds ratio have the advantage in comparison with the absolute risk that they depend less on the baseline risk. OR or RR are appropriate as an effect size measure when both variables are binary (cured vs. not cured). Conventionally, the odds ratio is commonly used in reporting treatment effects. OR is more or less equal to relative risk and so an odds ratio of 3 can be interpreted as 'the outcome of interest happens about thrice as often in the intervention group as in the control group'. But this is not accurate (simply equating OR and RR) as this will tend to overestimate the effect of treatment. But the level of this overstatement is almost always quite small, especially when the experimental and control event rates are less than 20%. However, for frequent events (approximately over 20%), the odds ratio turns out significantly greater than the relative risk, and falsely equating relative risk and odds ratio, then leads to an overestimation of the risk. For this reason, the relative risk is generally used in Cochrane reviews.



- *Effect sizes for continuous variables:* There are basically two effect size measures for continuous variables: <u>Simple difference between the mean values (DM)</u> keeps the original units and simply is the mean value of group X minus the mean value of group Y. <u>Standardized difference between the mean values (SMD):</u> This is carried out by dividing the DM by the pooled standard deviation of the two groups in the basic formula: SMD = (mean value of group X) mean value of group Y) □ pooled standard deviation of the two groups. Various formulas for the calculation of SMDs exist such as e.g. Cohen's d or Hedges' g, the latter avoiding a bias in Cohen's d, seen with small number of subjects.

Statistical methods for fixed vs. random effects:

Model	Statistical procedures	Measured effects
Fixed effects	*Mantel-Haenszel* (see SPMM notes on advanced statistics)	OR, Rate ratios, Risk ratios (RR)
	Peto	Approximates OR
Random effects	*DerSimonian-Laird*	Ratios and rate differences
None of the both above mentioned fixed effects methods allow for confounding to be assessed – hence they are not useful if confounding is suspected due to methodological issues. They are chi-square like tests, which need 2X2 tables for each individual study.		

A meta-analysis may not always give results that equate with large mega-trials that may be conducted later. For example google 'Lessons from an effective, safe, simple intervention that was not' (use of intravenous magnesium after heart attacks).

More about meta-analyses:

- Consensus statement on standard format for meta-analysis: **QUORUM** statement
- Consensus statement on standard format for RCTs: **CONSORT** statement
- **Meta-regression analyses**: Refers to a technique of regression wherein regression model is applied to meta-analysis to analyse which characteristics of the studies actually contributed to overall effect size.

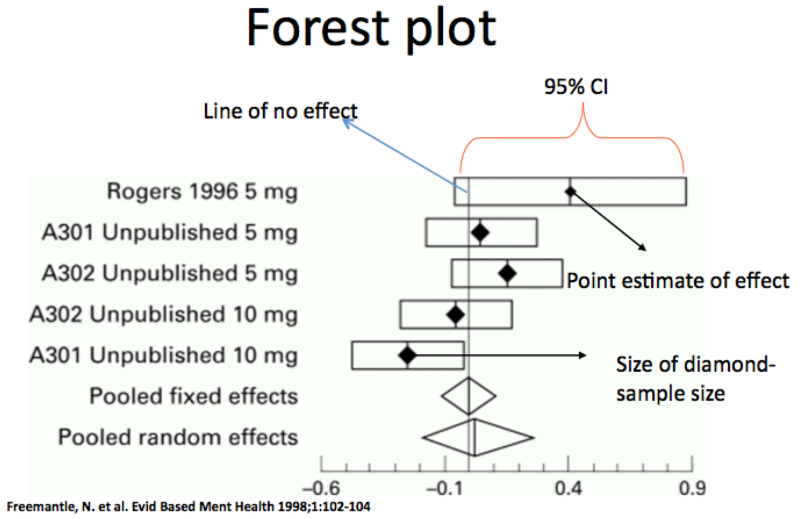


Figure 1 Effects of donepezil on quality of life (standardised effect size).

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Economic Studies

An economic analysis informs decision makers about cost effectiveness and benefits of an intervention. Such analyses are increasingly being used to decide upon service delivery. 5 major types of economic studies are tabulated here.

Type of study	Input / Output	Common usage	Summary (Hypothetical examples)
Cost Minimisation	**Input**: In terms of money (£££s) **Output**: not compared	Used when both interventions produce same outcome; hence just the costs are compared.	Which will cost me less? Buying CLOZARIL or buying ZAPONEX (another brand of clozapine) – £3.5 per tablet vs. £3.11 per tablet.
Cost effectiveness	**Input**: In terms of money (£££s); **Output**: Clinical measurement scales	Used when both interventions result in an outcome that can be measured using the same clinical scale.	Which is more cost effective? Does Clozaril give more economical improvement in psychotic symptoms or Zaponex? - £3000 per point improved in SAPS scale per patient per month vs. £3125 per point improved per patient per month.
Cost Utility	**Input**: In terms of money (£££s); **Output**: Life utility units e.g. DALY, QOL, etc.	Used when apart from the clinical effect one has also to consider holistically the disease burden, side effects and social impact, etc.	Which provides better value in terms of life utility? Does Clozaril give more life value for money spent than Zaponex? – £12500 spent per QALY gained for Clozaril, while £10500 spent for one QALY gained with Zaponex.
Cost benefit	**Input**: In terms of money (£££s). **Output**: In terms of money (£££s)	Used when two different interventions with different clinical outcomes are compared – in this case all outcomes are translated into monetary (£££) terms and are then compared. Often used by managers to decide how to spend the allocated money effectively.	Which is more beneficial for given cost – running a clozapine clinic or opening a service user's forum?
Cost consequences	**Input**: In terms of money (£££s) **Output**: In any natural units (days saved, number of deaths prevented, or QALYs etc.)	Used when two different interventions with different clinical outcomes are compared – as in cost-benefit or utility analysis – but instead of aggregating outcomes into a single common unit (money or QALYs), natural units are used which can then be translated according to the preference of stakeholders. Also called *disaggregated approach* as the outcomes are shown on a balance sheet of different dimensions rather than a single net value.	Which is more beneficial for given cost – running a clozapine clinic or opening a service user's forum? - But the benefits are decided in terms of preference values given to health states by stakeholders. Especially useful when health states are bound to change. Considers both health and non-health effects; both benefits and adverse outcomes can be considered by using a disaggregated approach. Unlike cost-benefit approach, all types of costs (including intangible ones) can be considered

How are the costs calculated?

There are various dimensions to the cost of an intervention.

Direct costs: Cost of hospital space, the cost of drugs, equipment, salary of doctors and other professionals, etc.

Indirect costs: Costs due to loss of productivity for the patient, the cost of having a caretaker for one's dog or a nanny for one's child when hospitalised, etc.

Intangible (not measurable) costs: Cost of pain and suffering, social stigma, physical discomfort, etc.

Opportunity cost: This refers to the cost of wasted opportunities – i.e. cost of losing an opportunity to attend a business meeting, having attended a dental appointment. Every action has an opportunity cost.

How are benefits calculated?

Economic benefits: Amount saved as a result of the prevention of illness and treatment, avoidance of admission, economic benefits of return to work, cost saved from quicker recovery.

Clinical benefits: improvement in rating scale scores, life years gained and deaths prevented.

The quality of life benefits: Improved general well-being, improved perceived life quality, regain of functions.

Health state preference values:

This refers to the value that a person will place in a health state in a hypothetical situation. There are four methods to calculate health state preferences;

1. **Rating scales:** Similar to the quantification of pain using a visual analogue scale, the respondent is asked to indicate along a line where would he place a particular health state. The zero point refers to death, and the maximum point refers to perfect health.
2. **Time trade-off measures**: How many of the remaining life years would a respondent spend in trying to get cured of a particular condition (in other words, how important is it for someone to get a complete relief from a health problem).
3. **Standard gamble measures**: Imagine there is a risky intervention for a health state that either cure if it succeeds or kills if it fails. How much risk would an individual accept to get cured the health problem with such an intervention? The measures are given as odds of success vs. odds of death – the respondents are asked to choose their preferred values.
4. **Willingness to pay (WTP):** How much people would be prepared to spend to prevent or cure a particular health state?

Quality indicators of economic analyses:

Several methodological aspects of an economic study must be critically appraised when determining the usefulness of a published economic analysis.

1. **Viewpoints:** The conclusions of an economic analysis could vary based on who is estimating the costs vs. benefits. The government and health ministry can have different views compared to the service users; the service users may have different views compared to public health bodies such as the WHO; the WHO may have a different view from doctors, etc. The conclusion of an economic study must be interpreted in the context of the viewpoints employed in the analysis.
2. **Clinical effectiveness:** The compared interventions may be cheap but what is the use if they don't work? It is unnecessary to compare two ineffective interventions to assess their relative economic benefits.
3. **Incremental values:** Consider two hypothetical interventions – using CBT and GET (Graded Exercise Therapy) for Chronic Fatigue Syndrome. Let's say CBT costs £1000 per course. GET costs £900 per course. CBT helps achieving 'recovery' in 5 out of 10 patients. GET helps 4 out of 10 patients.
   It is easy to quickly, but wrongly conclude that by spending an extra 100, pounds one is able to cure an extra patient – this may misleadingly appear to be highly cost effective. Consider the calculations below -

   Cost of treating 10 patients by CBT – £10,000
   As 5 get better, one spends 10,000/5 = 2,000 per patient recovered.

   Cost of treating 10 patients by GET = £9,000
   As 4 get better, one spends 9,000/4 = £2250 per patient recovered.

   So is it (£2250 - £2000) = £250 to achieve one additional recovery? No, this is also incorrect.

   As we spend £10,000 to get 5 persons better vs. £9000 to get 4 persons better, we are in fact spending a sum of £1000 extra for each additional recovery! This is because we can get one extra patient 'well' only if we treat a set of 10 with CBT. This incremental calculation must be done if two interventions are compared in an economic analysis (This computation is also called ICER – see below).

4. **Sensitivity analysis**: (please see the notes on meta-analyses for an introduction to sensitivity analysis). Various assumptions are usually made in economic analyses. For example, one assumes a standard rate of inflation and discounts the changes in foreign exchange values in the market when calculating costs. Also, a conventional 6% discount rate is assumed for year-on-year calculations. So it is pertinent to check the effect of the outcome of a reported economic analysis by systematically examining what will happen if the assumptions are changed in some way. This sensitivity analysis can be done in various ways;
   a. **One-way analysis**: Change one variable in a manner plausible at a particular economic circumstance to observe the effect on the outcome. This approach could underestimate the uncertainty as only one variable is changed e.g. only discount rates are changed.
   b. **Extreme scenario analysis**: Set one variable to either maximal or minimal values and examine the changes in the outcome. This approach invariably overestimates the uncertainty.
   c. **Two-way analysis**: Two variables are changed simultaneously to observe the effects e.g. inflation or discount rates and odds ratio favouring treatment. If more than two variables need simultaneous change this approach can turn out to be cumbersome; in this case, Monte-Carlo simulations are useful.

d. **Monte-Carlo simulatio**ns: The uncertainty in each utility is assumed to possess a probability distribution while estimating the predicted effects.

e. **Bootstrapping technique**: In this approach, instead of varying the observed parameters, the effect of resampling the data set is investigated. Conceptually, the RCT on which an economic analysis is based gets 'repeated' a large number of times. Any variability in outcome depends on the precision of the originally reported results.

The most popular economic evaluation method in health care is Cost Effectiveness Analysis. The results of a cost-effectiveness analysis can be summarised either as **average cost-effectiveness ratio (ACER)** which is equal to cost/effect or as **incremental cost-effectiveness ratio (ICER),** which is a measure of the additional cost per unit of health gained.

$$\text{ICER} = \frac{\text{The cost of proposed intervention} - \text{the cost of control intervention.}}{\text{The effect of proposed intervention} - \text{the effect of control intervention.}}$$

The incremental cost and incremental effect can be represented visually using the **incremental cost-effectiveness plane**. The horizontal axis divides the plane according to incremental cost (positive above, negative below) and the vertical axis divides the plane according to the incremental effect (positive to the right, negative to the left). This divides the incremental cost-effectiveness plane into four quadrants through the origin

Each quadrant has a different implication for the decision.

1. ICER falls in NORTHWEST: Positive costs and negative effects – no use of the intervention.
2. ICER falls in SOUTHEAST: Negative costs and positive effects – intervention is preferred.
3. ICER falls in NORTHEAST: Positive effects and positive costs – needs further consideration before acceptance.
4. ICER falls in SOUTHWEST: Negative costs and negative effects – not clinically effective - needs further study.

(Retrieved from http://www.biomedcentral.com/1472-6963/6/52)

Deciding 'value for money': In order to decide if an intervention offers real value for money, the ICER must be compared to the maximum amount that the decision-maker is willing to pay for health effects - λ (also called as **maximum acceptable ceiling ratio**). The intervention is deemed cost-effective if the ICER falls below this threshold and not cost-effective otherwise. For example, if a decision-maker is willing to pay £25,000 for a year of life, the intervention is considered cost-effective if the ICER is below £25,000 per life year gained. But note that the above plane does not take any uncertainty in the estimates of costs and effects into consideration. A decision maker will be interested to know how sure he/she can be in probabilistic terms to make such an investment. This probability can be identified from the incremental cost-effectiveness plane with reference to the decision maker's defined maximum acceptable ceiling ratio using **Cost-Effectiveness Acceptability Curve** - CEAC.

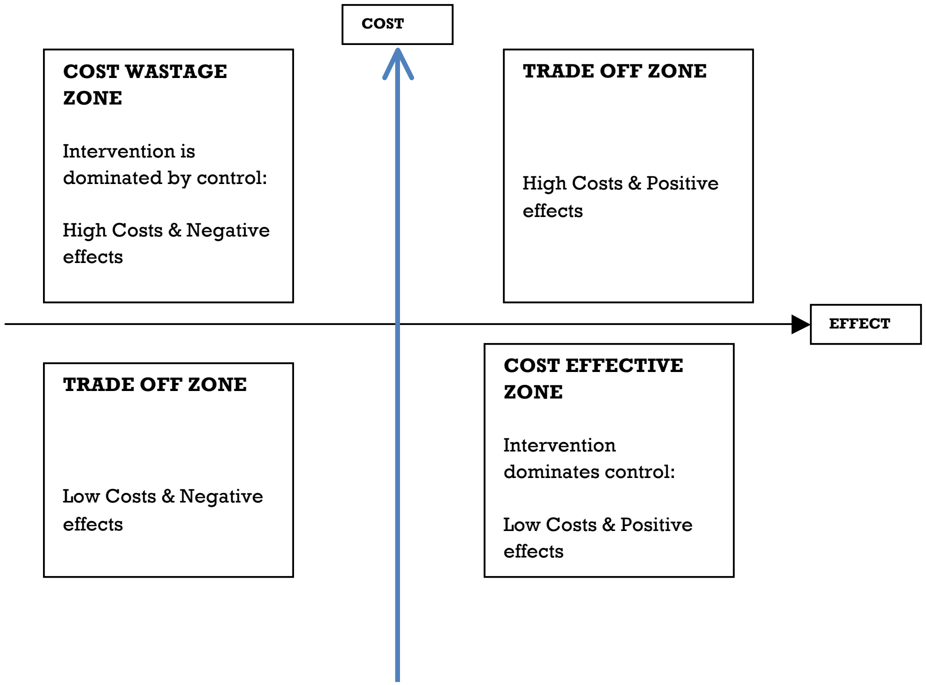
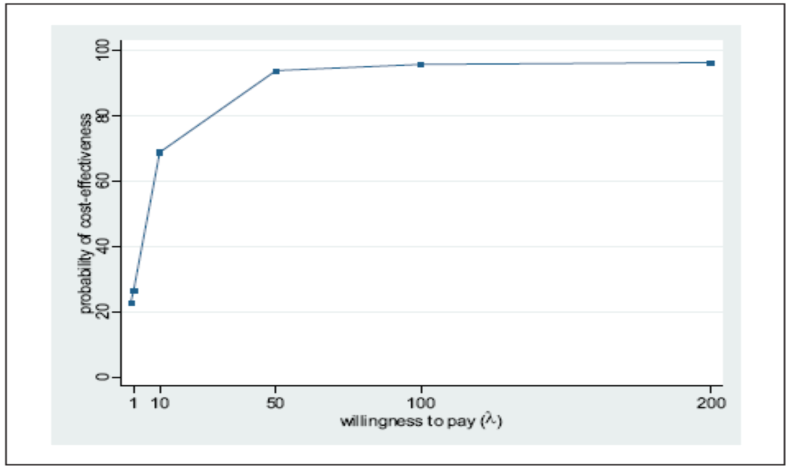


Figure 1 Cost-effectiveness acceptability curve



Cost effectiveness acceptability curve: The CEAC indicates the probability that an intervention is cost-effective compared to the alternative, given the observed data, for a range of λ (Willingness to Pay) values. The CEAC is a method of summarizing the uncertainty in estimates of cost-effectiveness.

Can we do a meta-analysis of economic analysis studies?

A basic method of summarising the results of multiple economic evaluations is the use of a narrative, descriptive summary. It is also possible to display different ICERs on a cost-effectiveness plane for a visual summary. **Nixon** proposed an alternative approach using a permutation matrix. In this approach, incremental effectiveness is categorised into positive, neutral or negative in each column. Incremental costs are categorised to higher, same and lower in each row. From this matrix, it is easy to pool multiple analyses and choose the most cost-effective intervention overall.

	Positive effect	**Neutral effect**	**Negative effect**
Higher costs	**No obvious decision**	**May reject**	**Certainly reject**
Same costs	**May accept**	**No obvious decision**	**May reject**
Lower costs	**Certainly accept**	**May accept**	**No obvious decision**

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